

CSET340

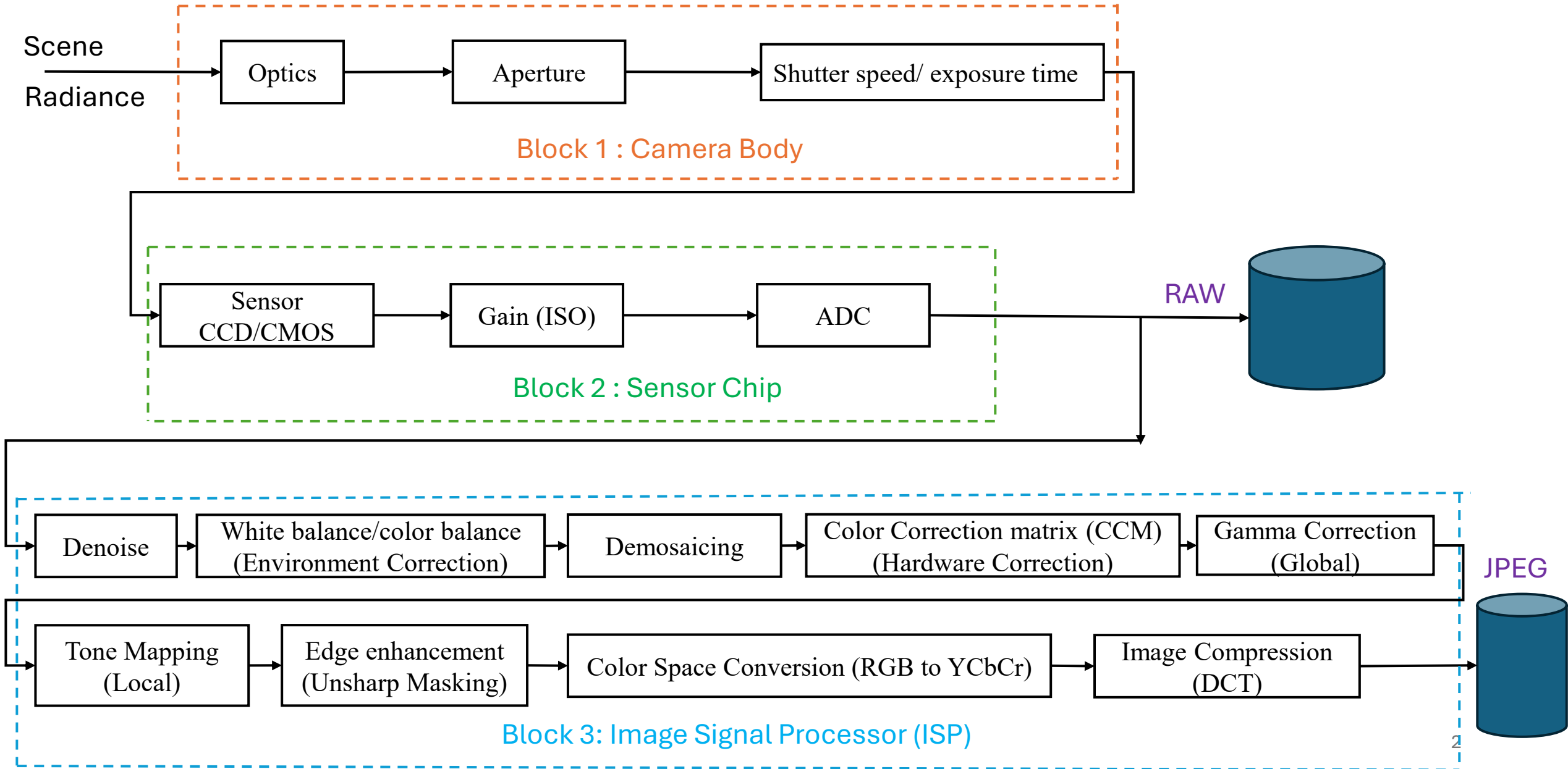
Advanced Computer Vision and Video Analytics

20th April. to 24th April. 2026

Overall Course Coordinator-
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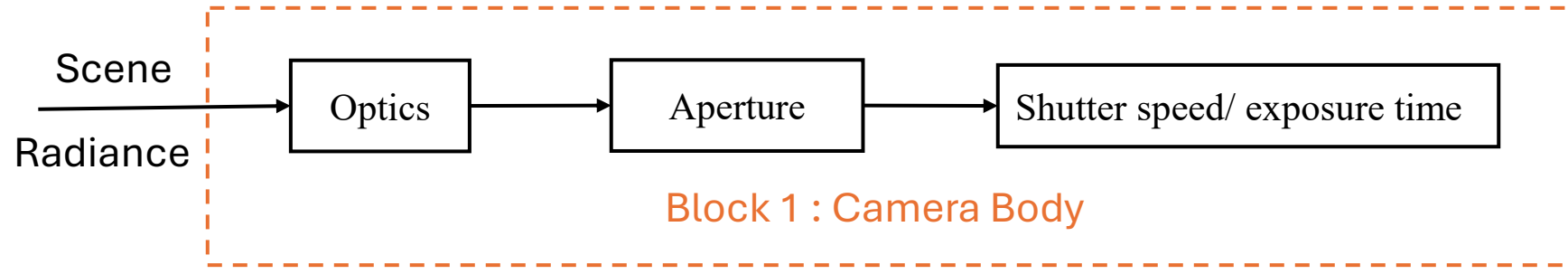
Note : Any query related to course then first connect with overall course coordinator.

Camera Image Sensing Pipeline: (Camera ISP) Overall flow diagram



Camera Image Sensing Pipeline: (Camera ISP) Overall flow diagram

Camera Body : It decides the input quality of the entire pipeline.



Once the light from the scene reaches to the camera, it must pass through the lens before reaching to the analog or digital sensor.

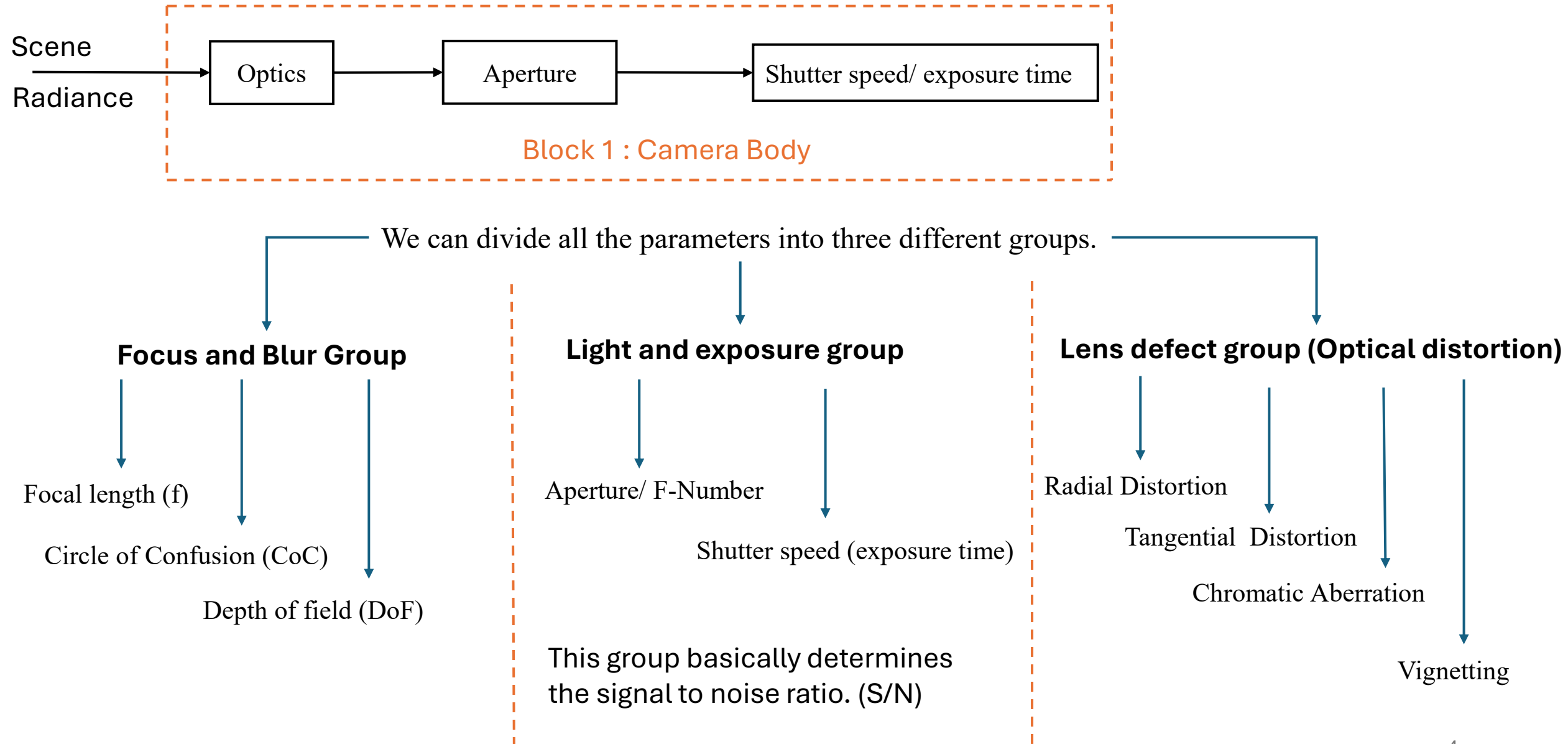
Irradiance is the total radiant power (P) incident upon a surface per unit area (A).

Radiance is the radiant power emitted, reflected, or received by a surface, per unit projected area, per unit solid angle.

Various factors that impact the output of the camera body –

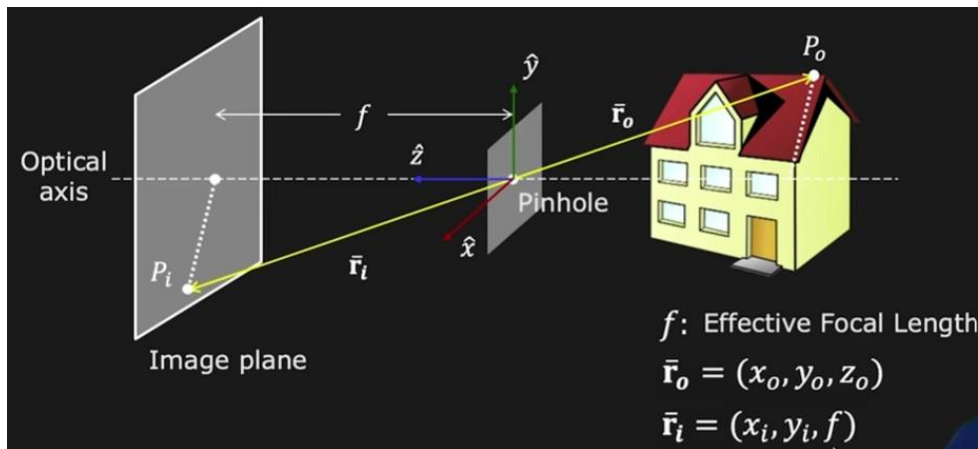
1. Focal length
2. Circle of confusion
3. Depth of field
4. Aperture
5. F-Number
6. Chromatic Aberration
7. Vignetting
8. Shutter Speed (Exposure time)

Camera Image Sensing Pipeline: (Camera ISP) Overall flow diagram



Focus and Blur Group: Focal length: Pinhole

Perspective projection of the pinhole camera



Optical axis: A axis perpendicular to the image plane

Focal length (f): Distance between pinhole and Image plane.

F-number = Focal length / diameter (fixed)

Here, F-Number is the dimensionless quantity that will tell you, how bright the system is relative to its focal length.

F-number	Aperture size	Light level	Depth of field
Low (f/1.8)	Large	Bright	Shallow (Blurry Background)
High (f/16)	Small	Dim	Deep (Everything is visible)

If focal length increases then image gets larger (zoom) but dimmer.
Why and How ?

Zoom

If you move the image plane (the sensor) Further away from pinhole the light rays have more distance to diverge.

The mathematical way –

$$H_i = H_0 \times f / D_0$$

H_i = Height of the image

H_0 = height of the object

F = focal length

D_0 = distance to the object

If f increase then H_i increase i.e image gets larger or zoom.

Dimmer

As focal length increase but the same amount of light is coming then this same amount of light has to spread over much larger surface area and image will become dimmer.

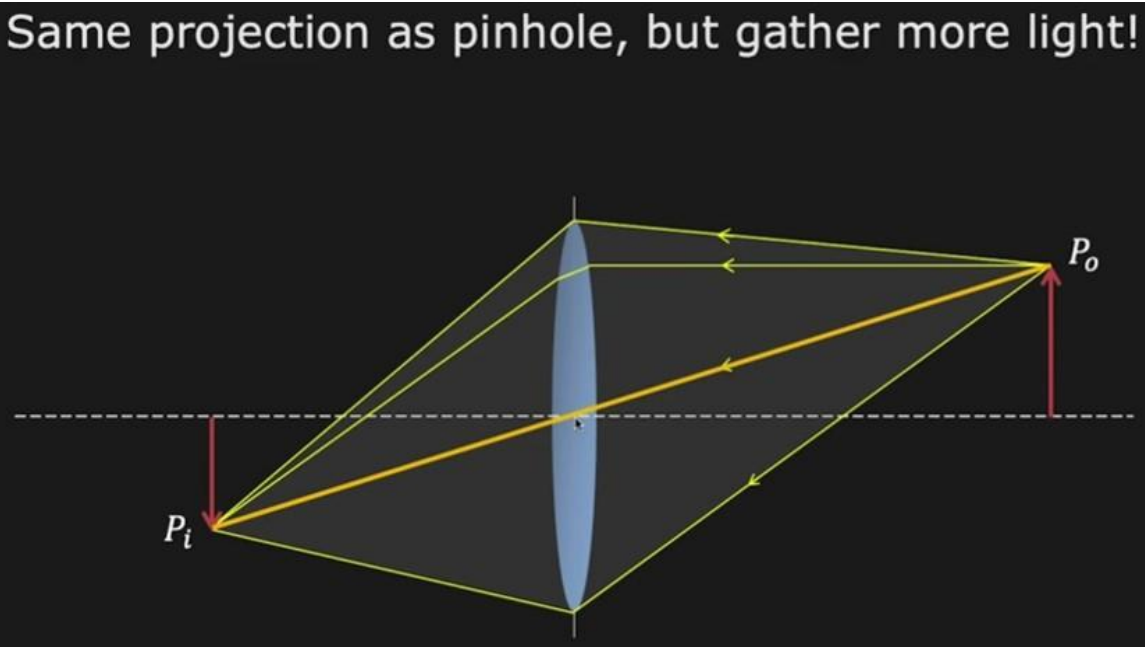
Inverse square law -

If you double the focal length ($2f$) then image becomes 4X larger in the area, it means it becomes 4 times dimmer.

Since the pinhole capture very little light. (With the **exposure time $T = 12$ sec**) To capture 1 frame you need to wait for at least 12 sec. Any computer vision system or mobile cameras can not wait for 12 sec to capture 1 single frame. (Less light so it will release less number of electrons so ultimately exposure time will be high)

Next is **Lens**

Focus and Blur Group: Focal length: Lens



In a lens, the focal length is a **fixed physical property** of the glass itself, determined by its **Bending Power**.

- The Physics:** The glass is curved and has a specific refractive index. This forces parallel light rays to converge at a specific point.
- The Definition:** The focal length is the distance from the lens to that convergence point (the focal point).

Scenario	Object Distance (u)	Focal Length (f)	Technical Result
Selfie (Close)	Very Small	Short (Wide)	High Perspective Distortion (Nose looks huge).
Landscape (Far)	Very Large	Short (Wide)	Low Magnification (Person looks like a dot).
Portrait (Far)	Medium/Large	Long (Tele)	Natural Perspective + High Magnification.

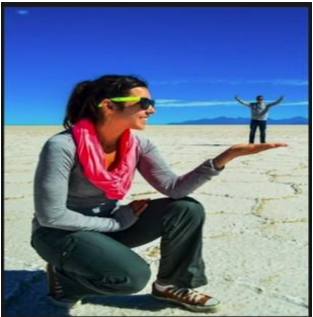
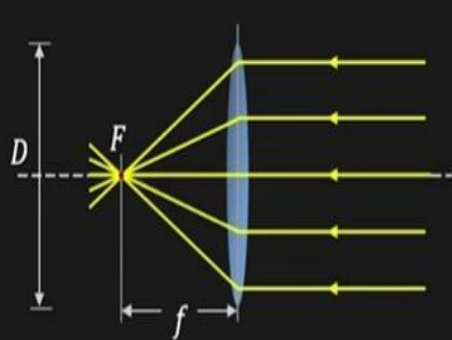


Image magnification is inversely proportional to the distance of object from the camera.

Aperture of the Lens and f-Number

Aperture of the Lens: It is the clear area of the lens that gather light from points in the scene.

Light receiving area of lens, indicated by lens diameter.



Aperture can be reduced/increased to control image brightness



D = Diameter of the aperture.

The **f-number** is defined as the ratio of the **focal length** of the lens to the **diameter of the entrance pupil** (the effective aperture opening).

f-number (f-stop, f-ratio) of Lens

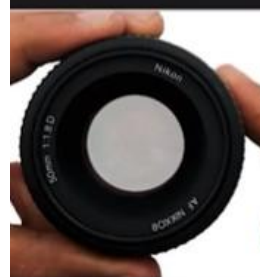
Convenient to represent aperture as a fraction of focal length

$$\text{Aperture: } D = f / N$$

$$\text{f-Number: } N = f / D$$

where **N** is called the **f-Number** of lens.

Ex: A 50mm focal length, f/1.8 lens implies:
 $N = 1.8$ ($D = 27.8\text{mm}$) when aperture is fully open



$N = 1.8$



$N = 4$



$N = 8$



$N = 11$

Numerical: F-Number and Aperture of the lens

- **F-number:** The ratio of the focal length to the aperture diameter. $N = f/D$
- **Aperture Diameter (D):** The physical width of the lens opening. $D = f/N$

Where:

- f = Focal length of the lens.
- D = Diameter of the entrance pupil (effective aperture).
- N = F-number (often written as f/N)

Problem: You have a telephoto lens with a focal length of **200 mm**. The effective diameter of the aperture is measured at **50 mm**. What is the f-number?

Step-by-Step Calculation:

1. **Identify variables:** $f = 200$ mm, $D = 50$ mm.
2. **Apply formula:** $N = 200/50 = 4$
3. **Notation:** This is expressed as **f/4**.

Problem: A standard portrait lens has a focal length of **85 mm**. If you set the camera to an aperture of **f/1.8**, what is the actual diameter of the lens opening?

Step-by-Step Calculation:

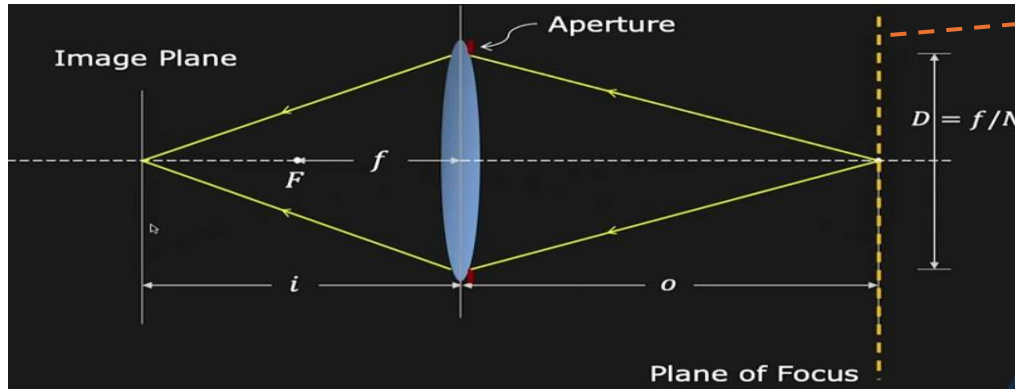
1. **Identify variables:** $f = 85$ mm, $N = 1.8$.
2. **Apply formula:** $D = 85/1.8$
3. **Result:** $D = 47.22$ mm.

• **Light Intensity:** A smaller f-number (like f/1.8) means a larger diameter (D). A larger diameter allows more light to hit the sensor, which is why "fast" lenses have low f-numbers.

• **Aperture Area:** Light gathering depends on the **area** of the circle (πr^2). If you change from f/4 to f/2.8, you are decreasing the f-number by a factor of $\sqrt{2}$ (Approx. 1.4), which effectively **doubles** the area and the amount of light entering the lens.

Lens Defocus.

Even though Lens gathers a lot of light but the price that we paid when we used it is, there is only one plane in the scene which is perfectly focused on the image plane.



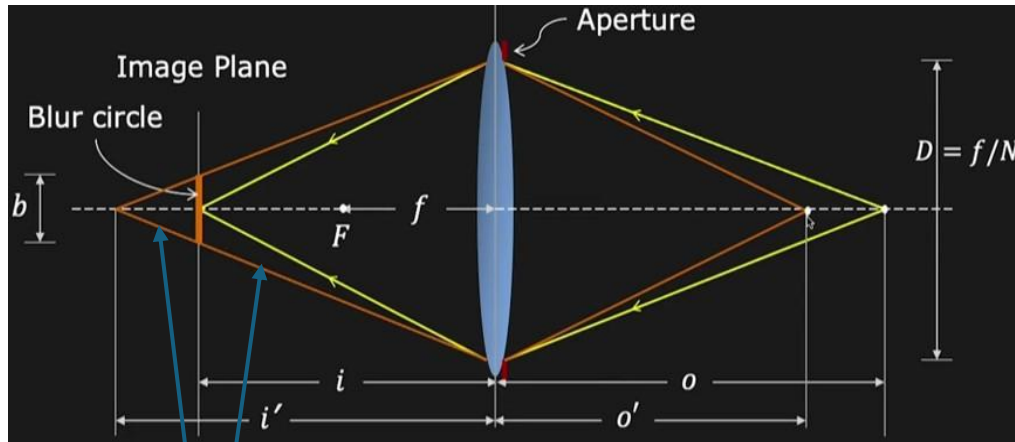
Plane of focus -

Here, this is the only one plane which is in focus. Any point on this plane is perfectly focused on the Image plane.

Here when the ray pass through it, it focuses on a single point in the image plane.

Now what happen if you lie outside this plane of focus.

Let's consider a point o' , as this point is closer to the lens its image is behind to the focused image plane.



These small and big triangle are the similar triangle

When the ray pass through it, it focuses not only a single point, but it will distribute over a circular disc on the image plane. **It is going to be blur and it is known as blur circle or circle of confusion.** Blur circle has a diameter equal to b .

From similar triangles:

$$\frac{b}{D} = \frac{|i' - i|}{i'}$$

Blur circle diameter:

$$b = \frac{D}{i'} |i' - i|$$

$$b \propto D \propto \frac{1}{N}$$

Blur circle diameter (b) is proportional to the diameter of the aperture (D) and Inversely proportional to the f-number of the lens.

Lens Defocus.

How to calculate Diameter of a blur circle

Blur Circle (Defocus)

Focused Point

$$\frac{1}{i} + \frac{1}{o} = \frac{1}{f}$$

$$i = \frac{of}{o - f}$$

Defocused Point

$$\frac{1}{i'} + \frac{1}{o'} = \frac{1}{f}$$

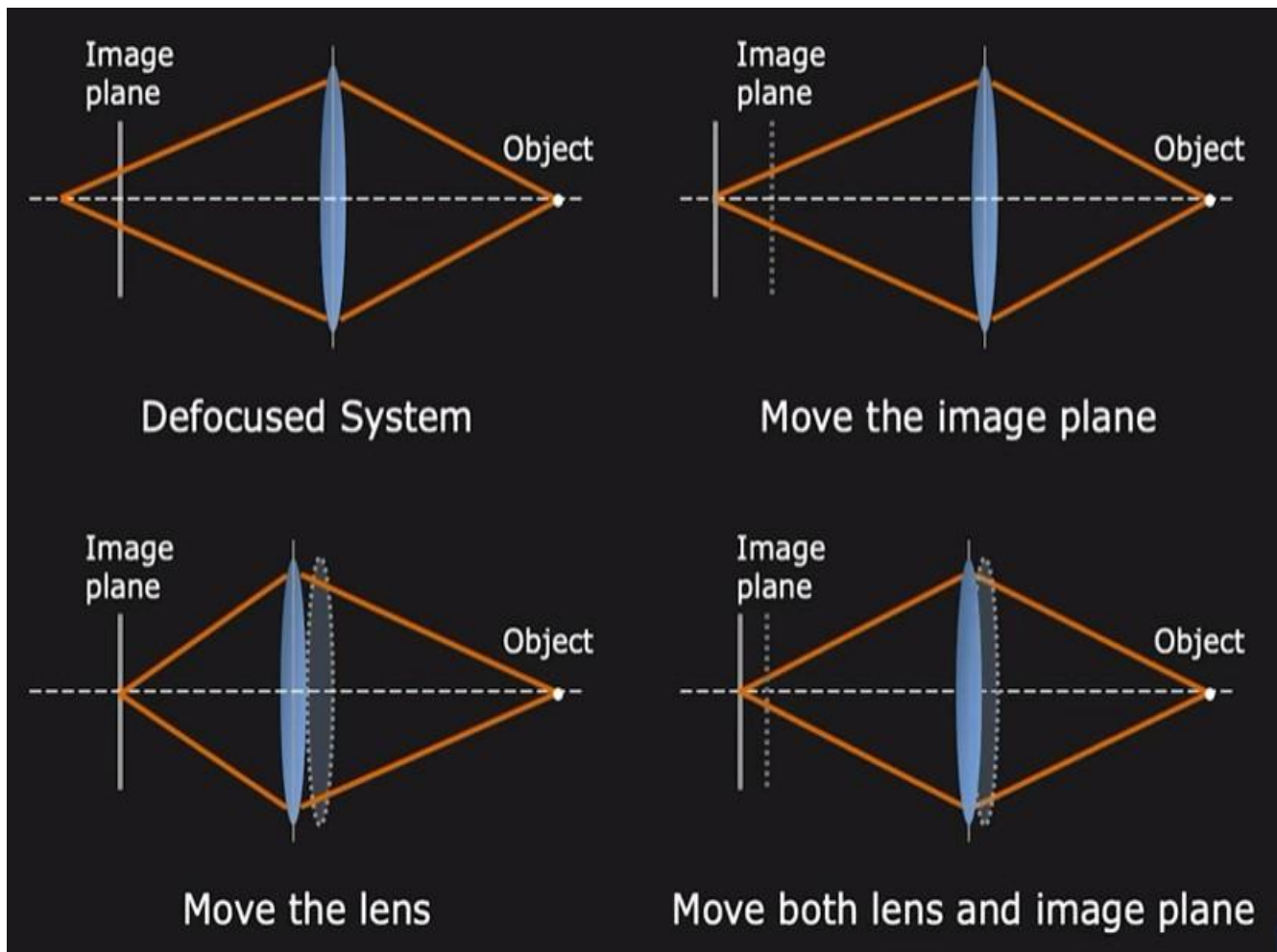
$$i' = \frac{o'f}{o' - f}$$

(Gaussian Lens Law)

$$i' - i = \frac{f}{(o' - f)} \cdot \frac{f}{(o - f)} \cdot (o - o')$$

$$b = Df \left| \frac{(o - o')}{o'(o - f)} \right| \quad b = \frac{f^2}{N} \left| \frac{(o - o')}{o'(o - f)} \right|$$

Focusing



Numerical: Calculating blur circle diameter in Lens

Problem: Calculating the blur circle (or **Circle of Confusion**) for a lens is more complex than a pinhole camera because it involves a lens that is actively focusing on a specific plane. The blur occurs when an object is located at a distance different from the one the lens is currently focused on.

Problem: You are using a **50 mm** lens set to an aperture of **f/2**. You have focused the lens on a person standing **2000 mm** (2 meters) away. However, there is a light source in the background **10,000 mm** (10 meters) away. Calculate the diameter of the blur circle created by that background light.

For a lens focused at distance P , the blur circle diameter c for an object at distance S is:

$$c = A \times \{f \times |S - P|\} / \{S \times (P - f)\}$$

- **$f = 50\text{mm}$**
- **$P = 2000$**
- **$S = 10,000\text{ mm}$**
- **$A = f/\text{f-number} = 50/2 = 25\text{ mm}$**

$$c = 25 \times \{50 \times |10,000 - 2000|\} / \{10,000 \times (2000 - 50)\}$$

Answer: The blur circle diameter is approximately **0.51 mm**.

Depth of field (DOF)

1. As there is one plane in a scene that is perfectly focused, all the objects are lie outside the range are going to be out of focus.
2. Degree of defocus will depend on the distance of it from the plane of focus.
3. Image is made up by pixel. Now, **the best focused points are those points where blur circle lies within single pixel.**

Condition for focused point or object = diameter of blur circle is less than or within the pixel size.

Depth of field: The range(min-max) of object distance for which the image is sufficiently well focused i.e the range over which the blur (b) is less than the pixel size is called depth of field of the imaging system.

or

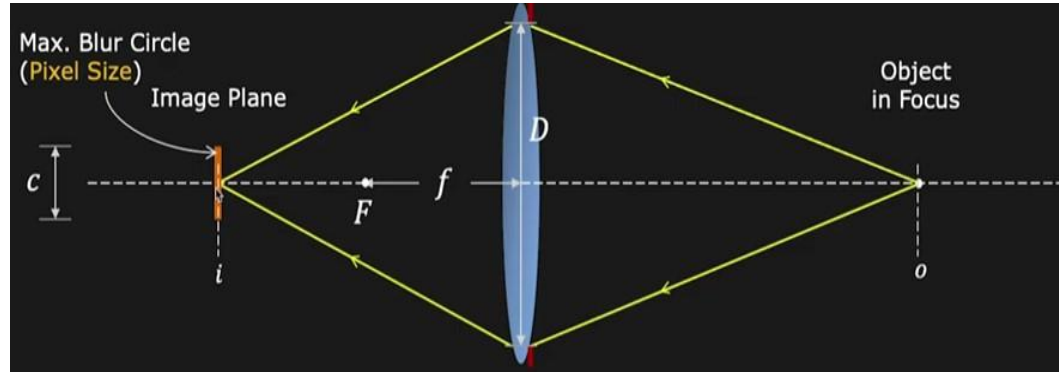
It is the distance between the closest and farthest objects in a picture that still look sharp and clear.



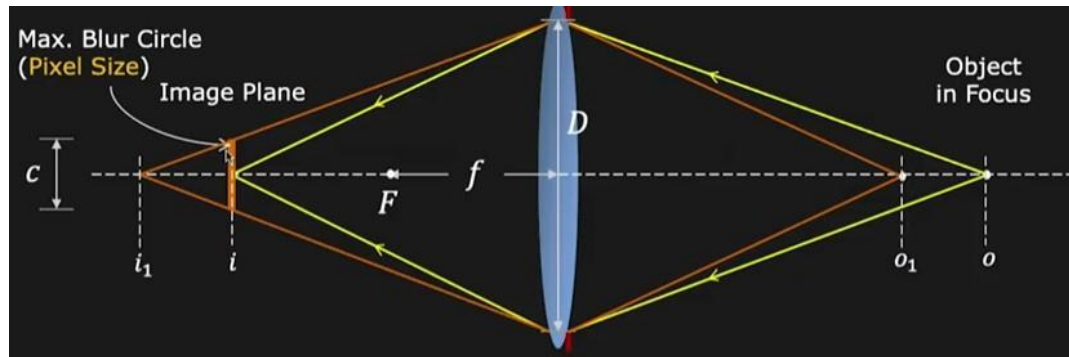
Depth of Field (DoF)

DOF for a lens System

1. First, define pixel size. let's call it C . Here O is the object which is in focus.

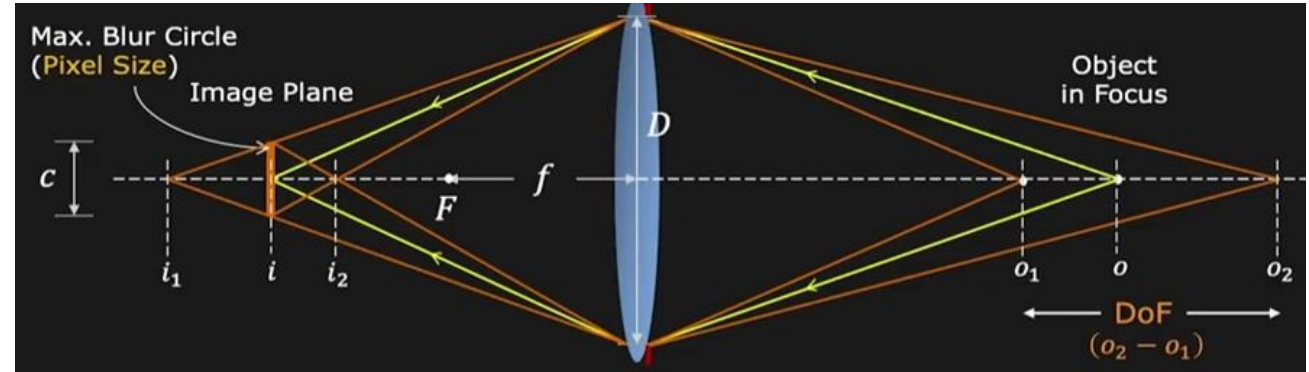


2. Now, we want to see what is the range of the Object (O) for which blur circle is smaller than C .



- 3.. Consider a point O_1 , for this point the blur circle diameter is exactly equal to the pixel size. Here, O_1 is close to the lens as compared to O , so the image is formed behind the image plane i.e. at i_1 .

4. Now, consider a point O_2 , where blur circle is exactly equal to the pixel. Here, O_2 is further away to the lens as compared to O , so the image is formed before the image plane i.e. at i_2 .



5. Depth of field (DOF) = $O_2 - O_1$.

If o_1 and o_2 are the nearest and farthest distances respectively for which blur circle is maximum c , then:

$$c = \frac{f^2(o - o_1)}{No_1(o - f)} \quad c = \frac{f^2(o_2 - o)}{No_2(o - f)}$$

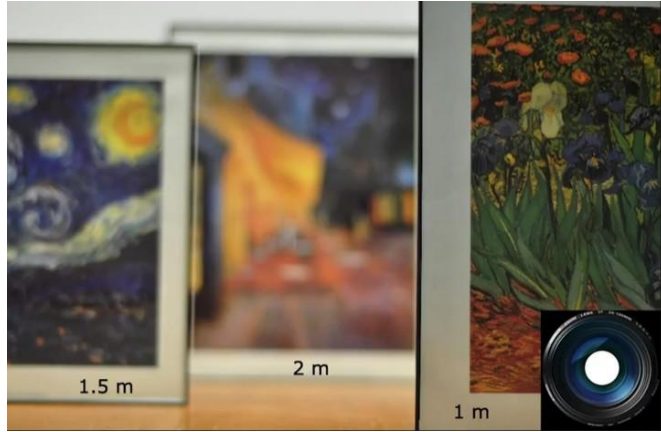
Depth of Field:
$$o_2 - o_1 = \frac{2of^2cN(o - f)}{f^4 - c^2N^2(o - f)^2}$$

Here c is the diameter of the blur circle..

Now, as the object is going closer to the lens then blurriness will increase, this brings us to the concept of hyperfocal distance. 13

Aperture Size: DOF Vs Brightness

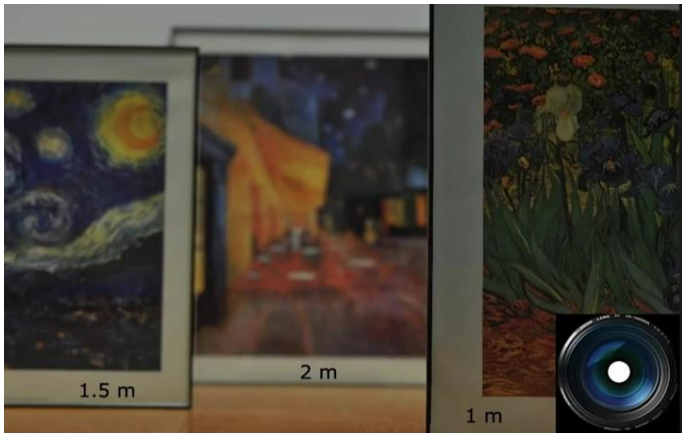
Here, we are using lens instead of pinhole camera because by using lens we can create the bright image.



Case 1: Consider focal length $f = 50\text{mm}$, Aperture $D = 25\text{mm}$ and $N = 2$



Case 3: Consider focal length $f = 50\text{mm}$, Aperture $D = 6.25\text{mm}$ and $N = 8$



Lets consider 3 different images at different distance from lens. Where lens is focused at 1m distance.

Observations

Large Aperture (Small f-Number)

- **Bright** Image or **Short** Exposure Time
- **Shallow** Depth of Field

Small Aperture (Large f-Number)

- **Dark** Image or **Long** Exposure Time
- **Large** Depth of Field

Case 2: Consider focal length $f = 50\text{mm}$, Aperture $D = 12.5\text{mm}$ and $N = 4$

Shutter speed (Exposure time)

Shutter speed (exposure time).

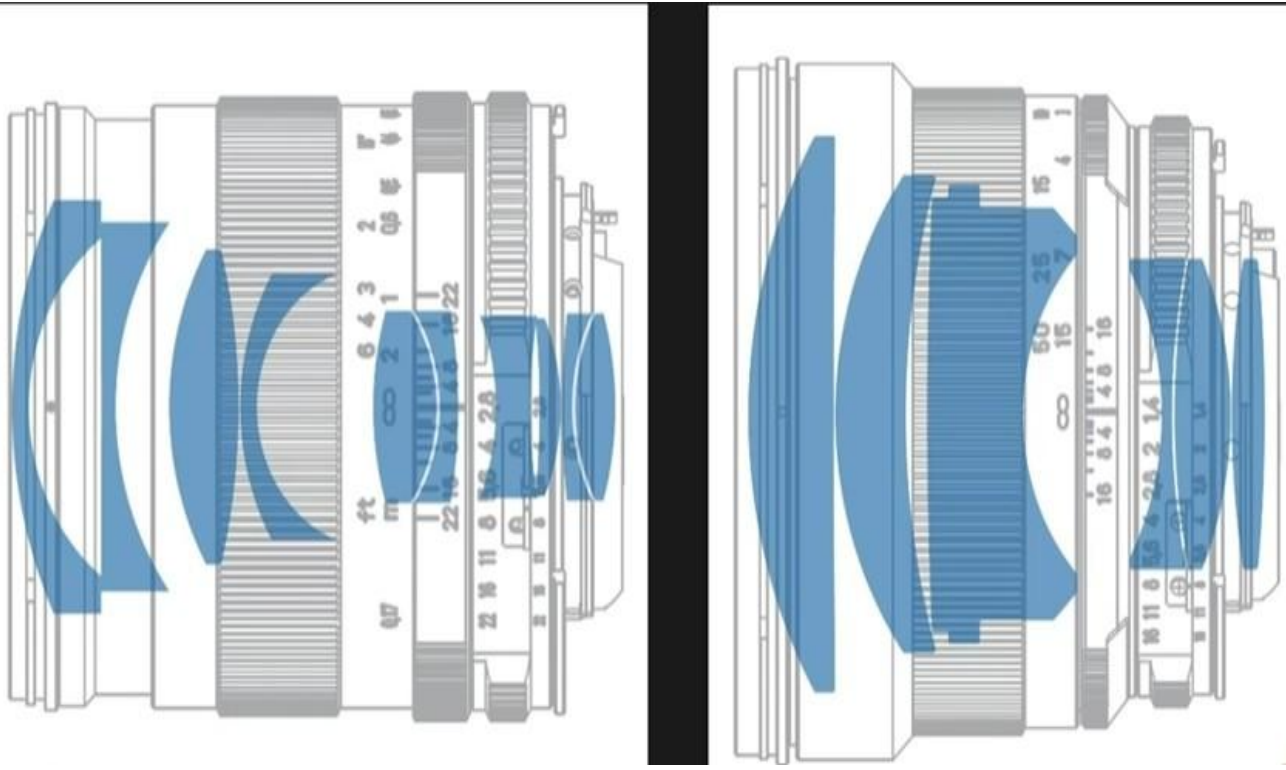
The shutter speed directly controls the amount of light reaching to the sensor.

It determines, if images are underexposed or overexposed.

Metric	Fast Shutter (Low Exposure)	Slow Shutter (High Exposure)
Photon Count	Low	High
Signal-to-Noise Ratio	Low (High Noise)	High (Low Noise)
Motion/Shake Risk	Low (Sharp)	High (Blurry)

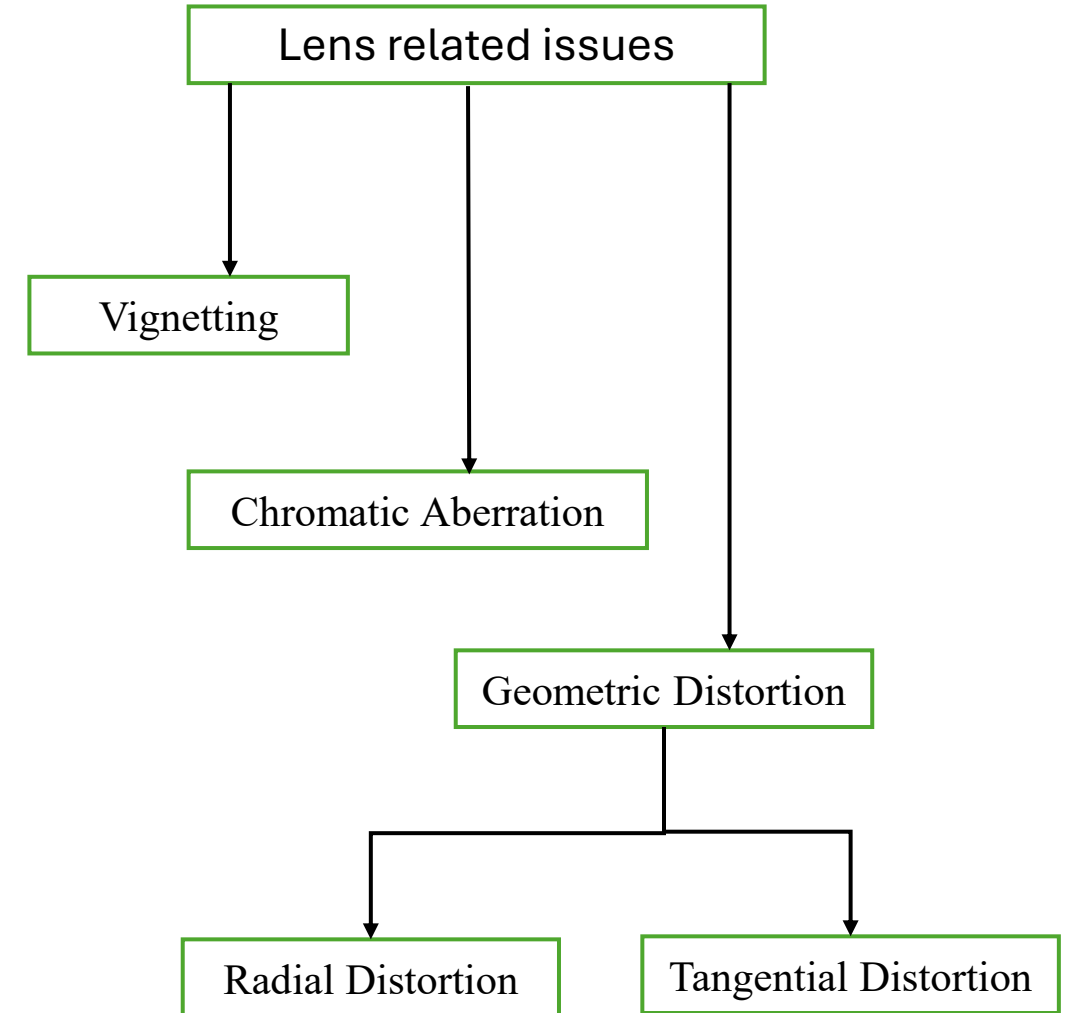
Signal to Noise ratio (S/N) = Actual Signal / Noise S/N ratio should be high or low ?

Lens related issues:



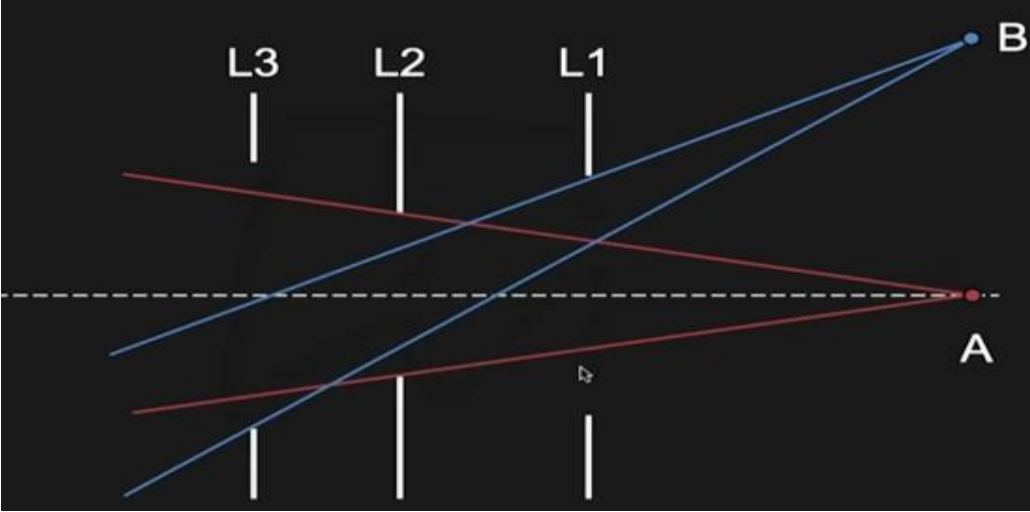
Till now we always talked about a single lens but practically, its compound of lens. Compound lens are used to compensate the undesirable effects of each other, so that quality can-not be compensated.

Design of lens = Art + science



Lens related issues

Vignetting

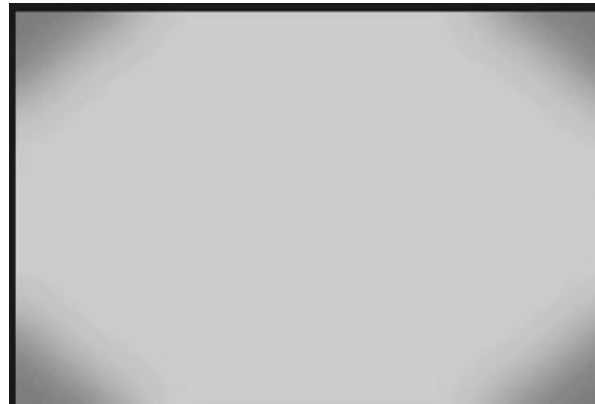


Here, we have three different lenses (L1, L2, L3) with different size and opening that allows different amount of light pass through it.

When you place a point on the optical axis i.e. point A, the amount of light that is reaching to the sensor is more as compared to point B (which is a point away from optical axis). Here, B is the point at the boundary of the lens.

Due to this, the edges or corners of an image appear darker or less saturated than the center.

Darkening or fading of an image's corners.



Brightness fall-off (Vignetting)
in image of a White Wall

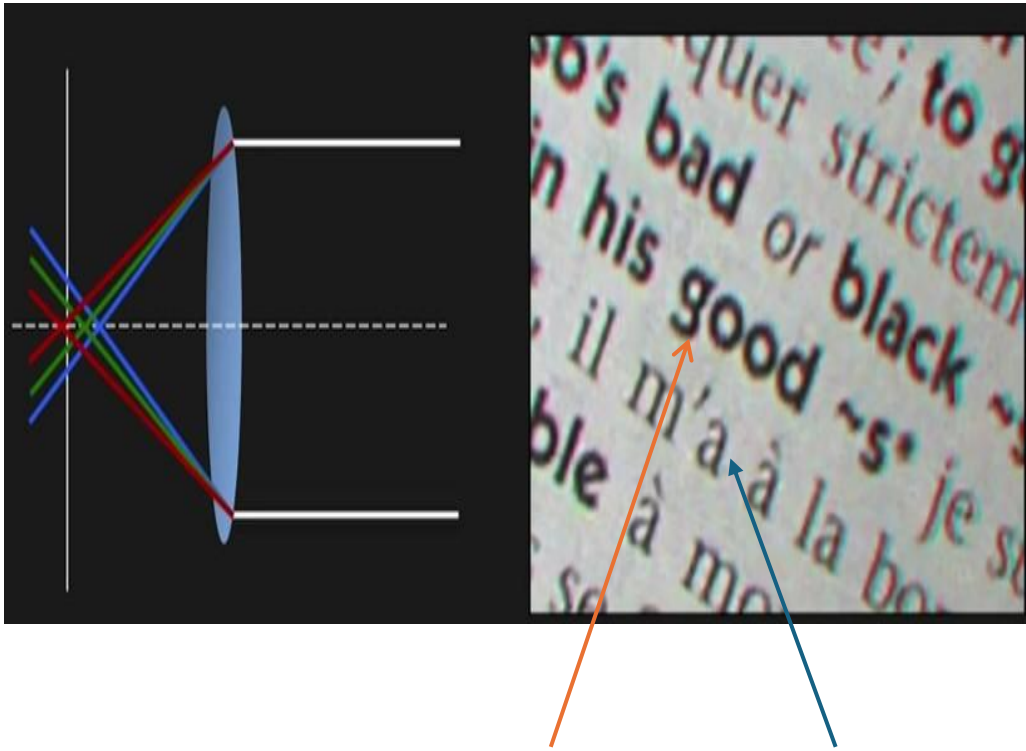


Brightness fall-off (Vignetting)
in image of a Natural Scene

Lens related issues

Chromatic Aberrations (Deviation)

When a lens fails to focus all colors of light to the same point. In a photograph, this looks like "color fringing".



Here, there is change in color

Lens made by certain material like glass or plastic.



This material has some refractive index.



Refractive index is function of wavelength

Refractive index = $f(\text{wavelength})$



Since R,G,B has different wavelength. Then, when white light is coming to the sensor then we can see blue bends more as compared to red.



It directly implies blue color is focused at different point as compared to red color.



As a result, there is some shift in the color in image that is noting but chromatic aberrations.

Geometric Distortion:

It is an error of **position**, not focus. The image can be perfectly sharp, but the objects in it are the wrong shape or in the wrong place.

Radial Distortion

It occurs when light rays bend more (or less) as they move from the center of the lens toward the edges, causing the magnification to vary across the frame.

Barrel Distortion

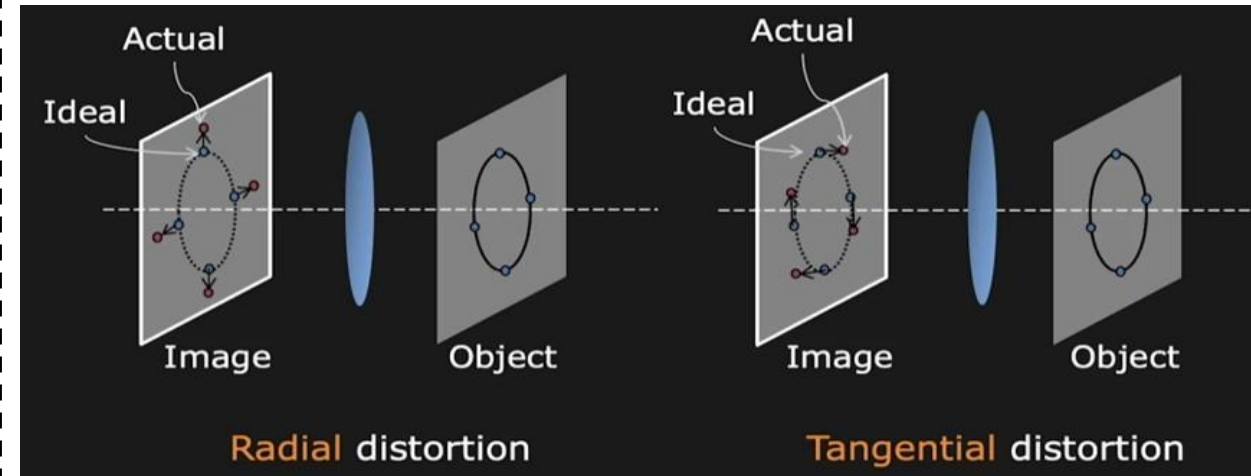
Magnification decreases toward the edges. Straight lines bow **outward**,

Tangential Distortion

Tangential distortion occurs when the lens assembly is not perfectly parallel to the image sensor., some parts of the image appear closer (more magnified) and others further away than they should be.

Pincushion Distortion

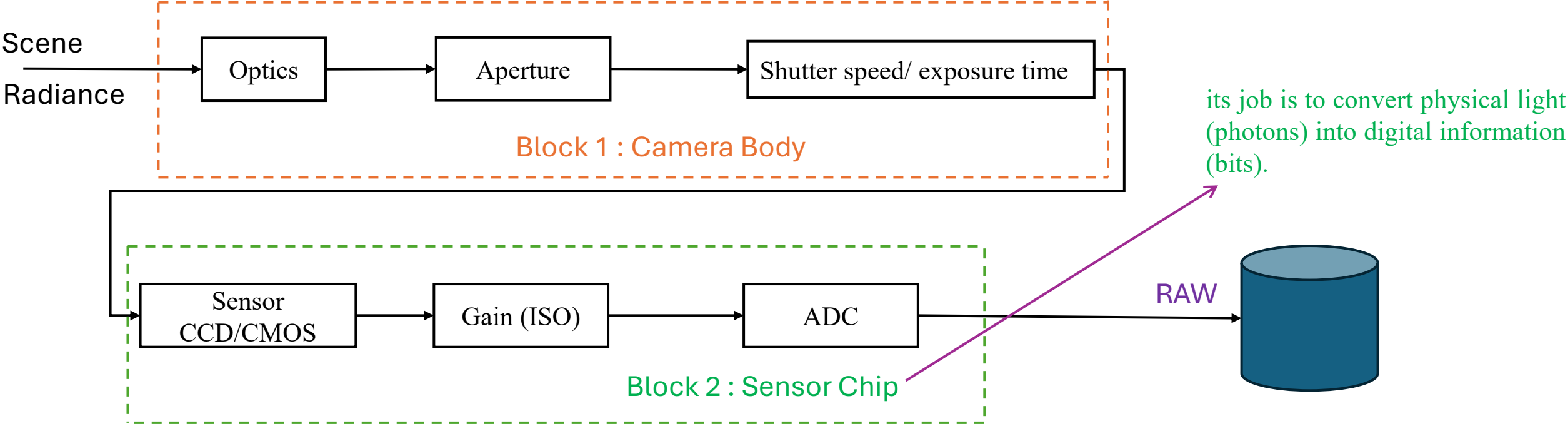
Magnification increases toward the edges. Straight lines bow **inward** toward the center.



Straight lines in the scene is not projected as straight line on the image plane.

As you move away from the centre of the image the twisting will increase.

Camera Image Sensing Pipeline: (Sensor Chip)



Sub-Block	Input	Output	Responsibility	Various factors that impact the output of the Sensor Chip –
CCD/CMOS	Photons	Electrons (Charge)	Convert Light to Electricity.	1. CCD/CMOS
Sensor Gain	Weak Voltage	Strong Voltage	Boost the signal for processing.	2. Sampling Pitch
ADC	Analog Voltage	Digital Bits (RAW)	Convert electricity to numbers.	3. Fill Factor
				4. Chip size
				5. Analog Gain
				6. Sensor Noise
				7. Analog to digital conversion (ADC)

Sensor CCD & CMOS: Image Sensing:

Converting light into electrical charges

Image sensors are made by silicon.



When silicon atom hit by photon of sufficient energy.
It releases electron and a electron hole pair is generated.

It means silicon does most of the work for you.



The challenging task is to read out these electrons and convert them into voltage. It will become more challenging when millions of pixel were present.

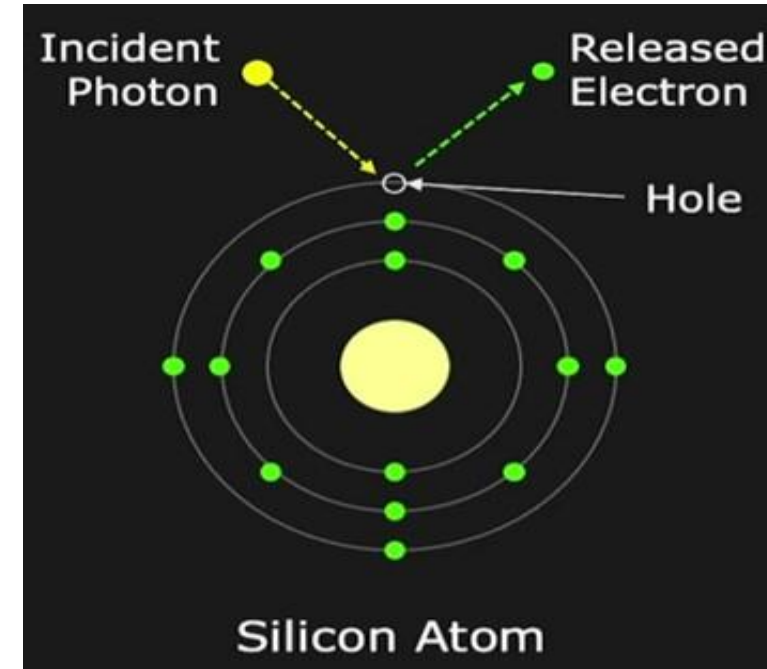
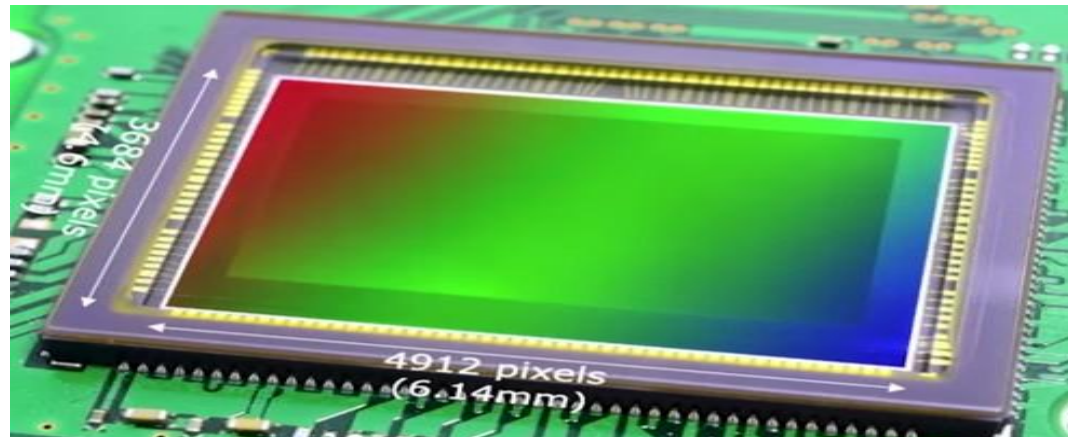
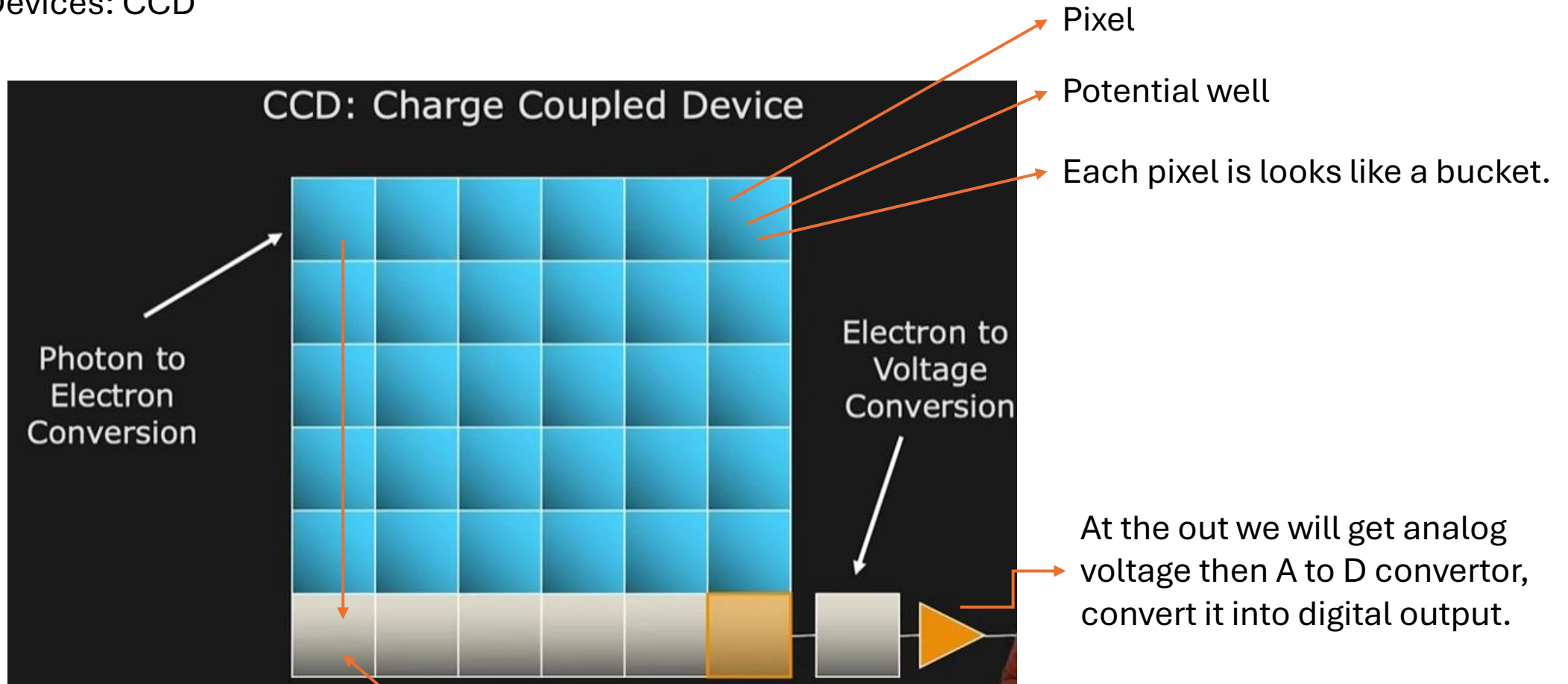


Image Sensor



Sensor Chip: CCD

Charge Couple Devices: CCD

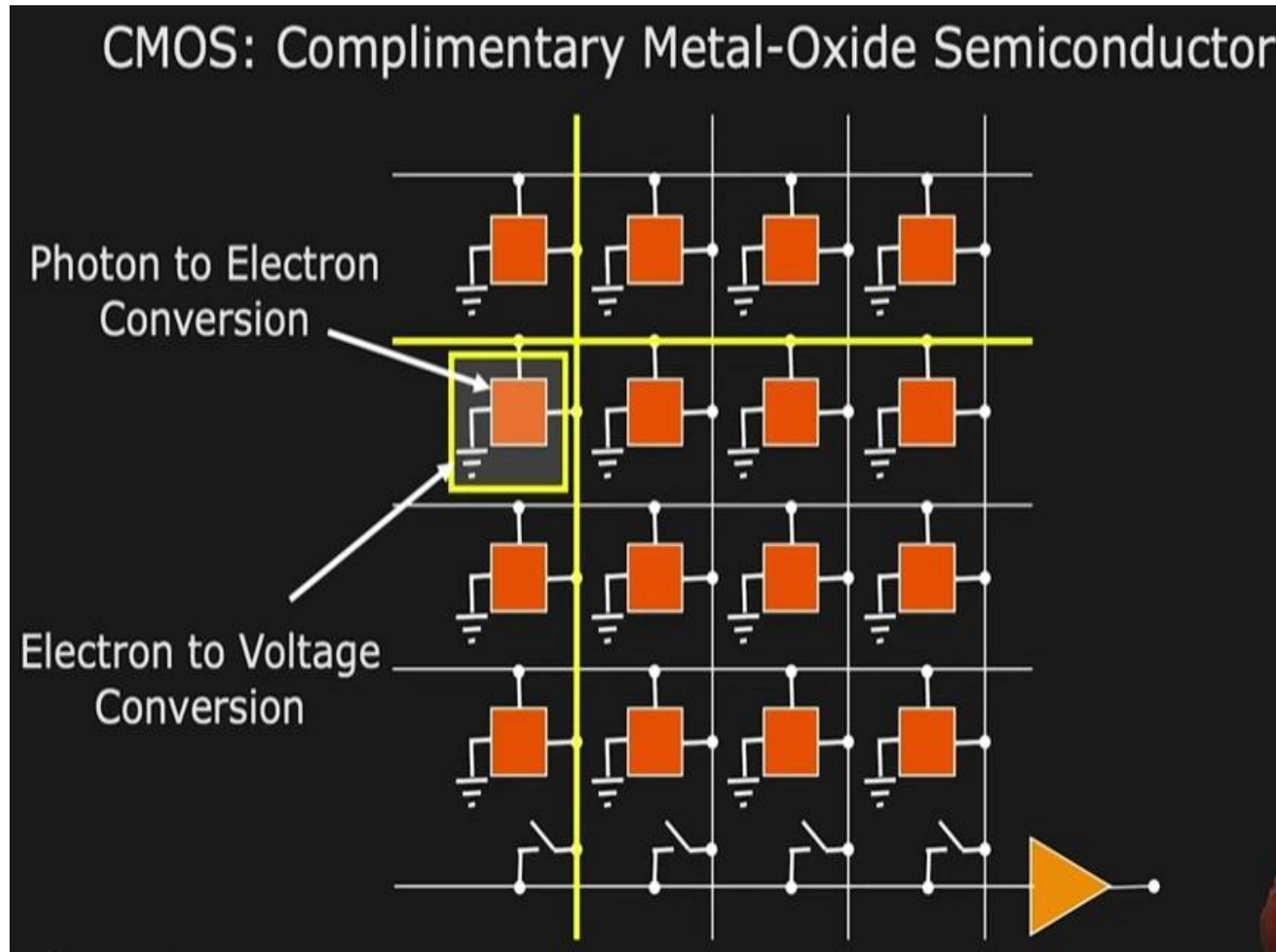


Each Row passes its electron to the next row. It called as bucket brigade.

Electric field applied beneath these bucket such that it will help to pass the electron row by row.

At bottom row, all the electron are collected and Here, total electron presented in each pixel is converted into voltage one by one.

Sensor Chip: CMOS



Here each pixel has its own circuit while in CCD, one circuit is shared by the entire chip.



Next to photon to electron conversion there is a circuit which converts electron to voltage.



You can select any pixel and find its voltage.



It may be helpful when someone is interested into a particular region.



As the circuit is near to the sensor, a price need to Pay, which is total light sensitive area is smaller.

Sensor Chip: Sampling pitch & Fill Factor

Sampling Pitch or Pixel Pitch

Sampling Pitch is the physical distance between the centers of two adjacent pixels on an image sensor. (It is typically measured in micrometers.)

It basically represents the **spatial sampling frequency** of the camera.

Smaller sampling pitch $\longrightarrow$ Higher pixel density $\longrightarrow$ Higher resolution for a given active chip area.

Fill Factor is a critical metric that defines the efficiency of the **CCD/CMOS** block. It is the ratio of the light-sensitive area of a pixel to its total area (which is determined by the sampling pitch).

Fill factor = (Active photodiode area / Total pixel area) x 100

A pixel is not just a "bucket" for light. Each pixel requires support circuitry to function:

Transistors: For resetting the pixel and reading the signal.

Metal Wiring: To transport the electrical charge to the **Sensor Gain** and **ADC** stages.

Isolation Trenches: To prevent **Crosstalk** between neighboring pixels.

These components take up physical space on the silicon, leaving less room for the actual photodiode.

Fill factor is a primary driver of the **Signal-to-Noise Ratio (SNR)**:

Low Fill Factor: Most of the photons hitting the sensor land on "dead space" (wires/transistors) and are lost. This results in a weak signal and high **Shot Noise**.

High Fill Factor: More photons are captured, leading to a "cleaner" RAW image with higher SNR.

Sensor Chip: Chip size and analog gain

Chip Size: It is the total physical area of the sensor.

Larger chip is essentially a larger "bucket" for light, which drastically improves the **Signal-to-Noise Ratio (SNR)**

If Larger chips (Like in Digital single lens reflex **DSLR** camera)

Fewer number of chips packed into each wafer.

Probability of chip defects goes exponentially with the increment in the chip area.

Elseif Smaller chip size.

More number of chips are packed into each wafer.

as size is less than there is very less probability of chip defect.

Analog Gain:

It is the process of increasing the voltage of the electrical signal generated by the pixels before it is converted into a digital number.

When photons hit a pixel, they create a very tiny electrical charge (electrons). In low-light conditions, this charge is so weak that it might be lost or misread by the **ADC**.

The Amplifier: The Analog Gain block is a physical amplifier circuit. It takes that weak voltage and multiplies it.

The Goal: To "stretch" the signal so it occupies the full range of the **ADC**. If you have a 12-bit ADC (0 to 4095), you want your brightest pixels to be near 4095, even if the actual light was very dim.

Problem: Higher ISO means - higher gain – It allows the gain to perform better under low light condition., **But it amplify noise also.**

Camera ISP:

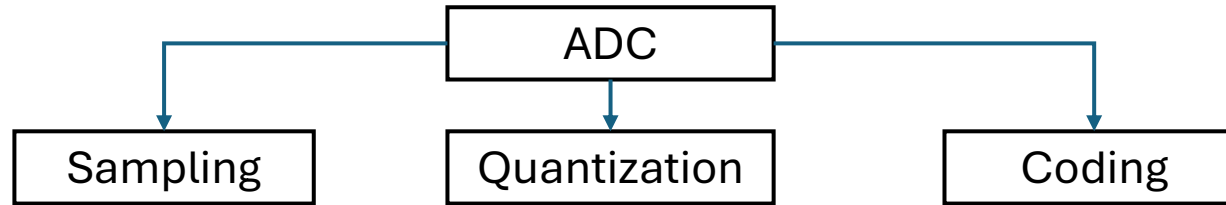
The ISO number (like ISO 100) is a **Standardized Rating** of sensitivity.

- **Base ISO (e.g., ISO 100):** This is usually **Unity Gain (1x)**. The electrical signal is not amplified at all.
- **ISO 200:** This is exactly **double** the sensitivity of ISO 100. To achieve this, the analog amplifier multiplies the voltage by **2x**
- **ISO 400:** This is double ISO 200. The amplifier multiplies the voltage by **4x**.

ISO Setting	Gain Multiplier (Approx)	Change in "Stops"
ISO 100	1x (Base)	0
ISO 200	2x	+1 Stop
ISO 400	4x	+2 Stops
ISO 800	8x	+3 Stops
ISO 1600	16x	+4 Stops
ISO 3200	32x	+5 Stops

Sensor Chip: Analog to Digital Convertor (ADC)

The **Analog-to-Digital Converter (ADC)** is the "bridge" where your data finally leaves the world of physics (electricity) and enters the world of mathematics (bits).



Step 1. Sampling: The ADC cannot process a continuous stream of electricity. It must "take a snapshot" of the voltage at a specific moment in time.

In a Sensor: The ADC samples the voltage level from each individual pixel's capacitor.

Step 2. Quantization (Assigning a Level)

The continuous analog voltage is mapped to a discrete "step" on a digital scale.

• **The Scale:** This is defined by the **Bit Depth**.

- **10-bit:** 1,024 possible levels.
- **12-bit:** 4,096 possible levels (Standard for most **smartphones**).
- **14-bit:** 16,384 possible levels (Standard for DSLRs).

• **Quantization Error:** Since the voltage might fall *between* two steps, the ADC has to "round" to the nearest value. This rounding error is a form of noise!

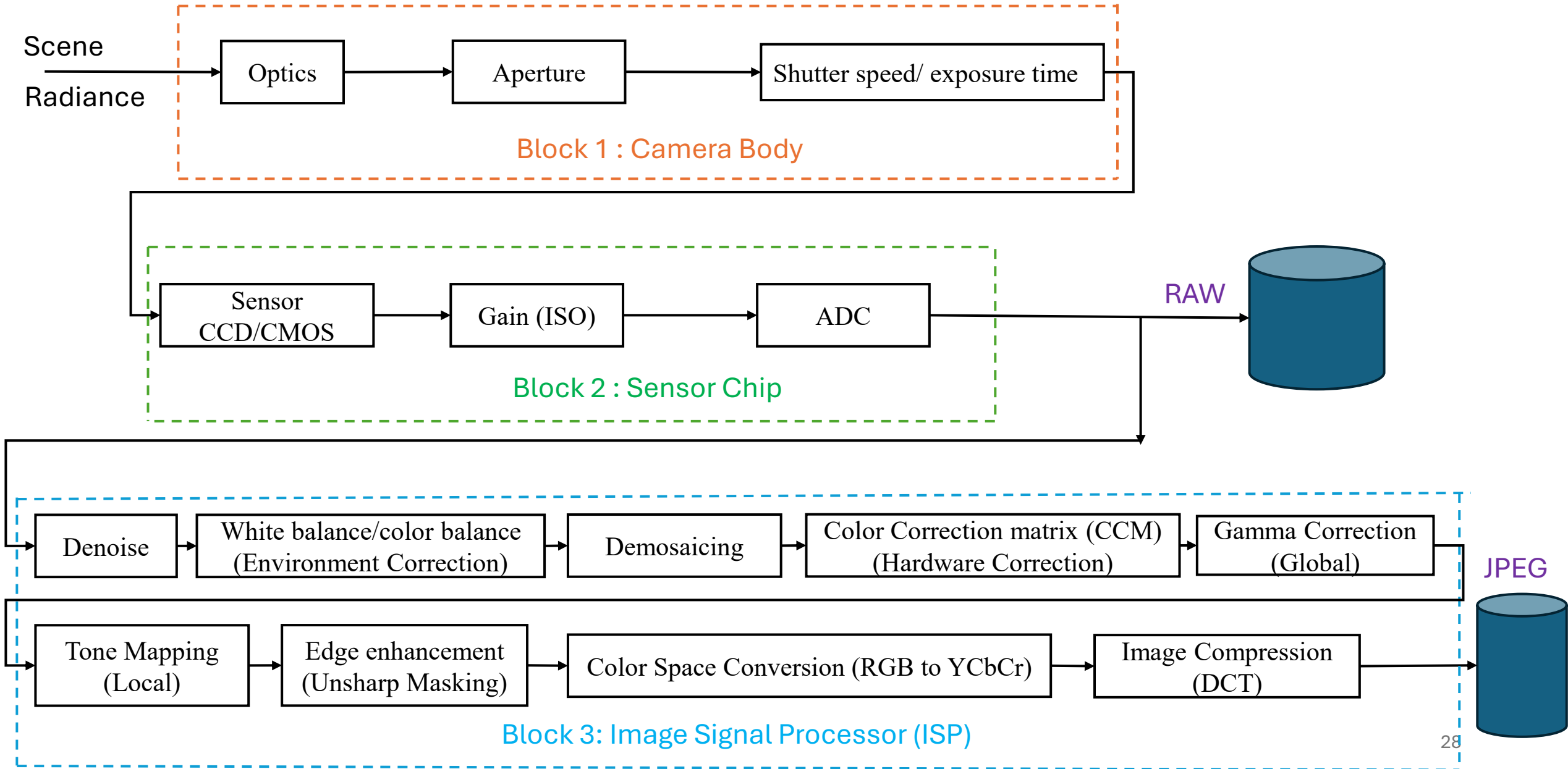
Step 3. Coding (Converting to Binary)

Once a level is assigned (e.g., Level 3,042), the ADC converts that number into a binary string (0s and 1s).

• **Output:** This binary string is what makes up the **RAW Bayer Data**.

• **Integrity:** At this stage, the data is relatively safe from external interference. Unlike an analog signal, a digital "1" is hard to mistake for a "0."

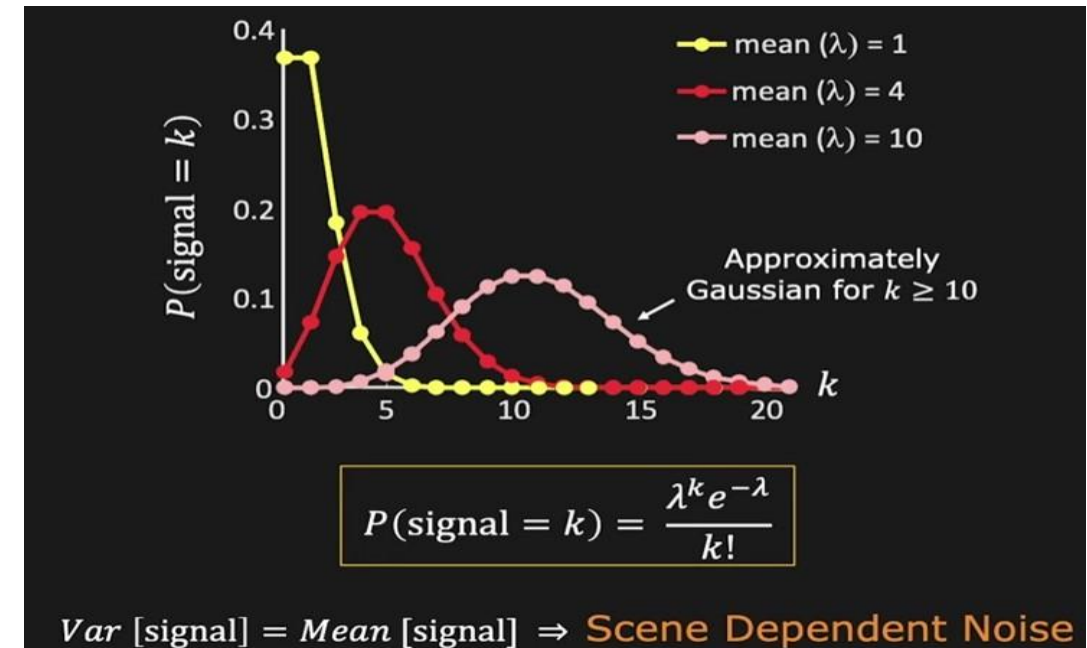
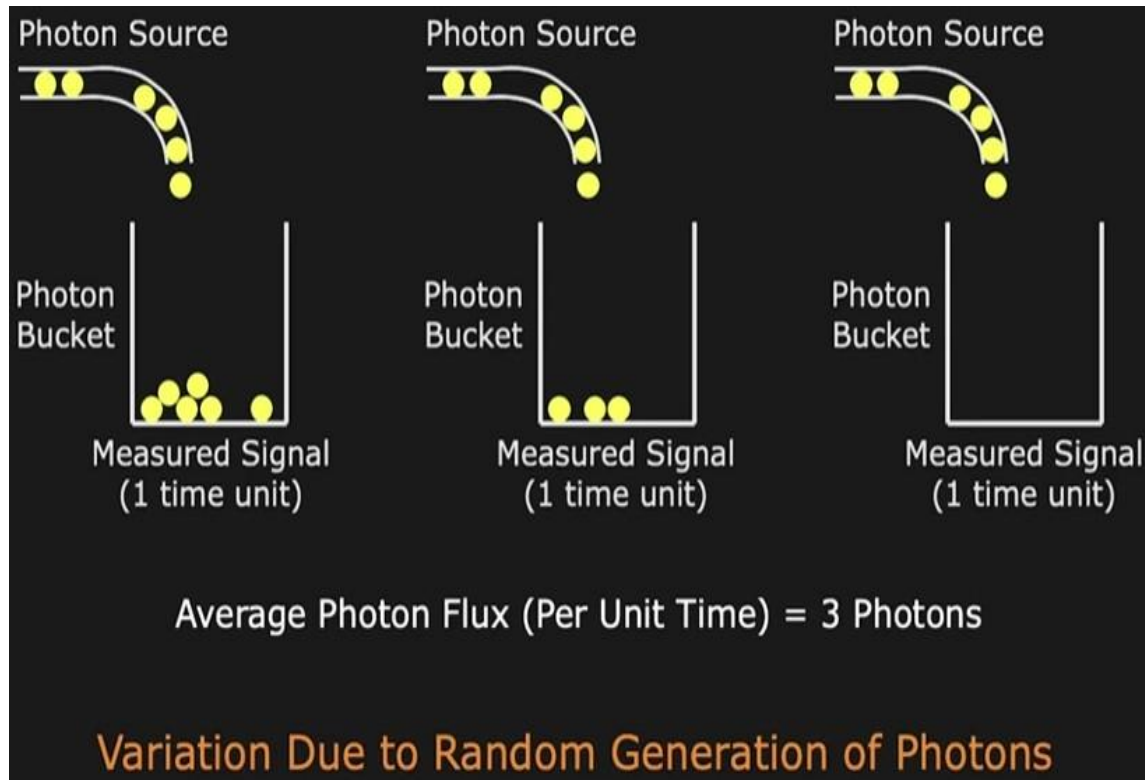
Camera Image Sensing Pipeline: (Camera ISP) Overall flow diagram



Denoising: Different Noise and their denoising approaches

Photon Shot Noise (Scene dependent noise)

1. It is due to random or quantum nature of the light itself.
2. Photons are arriving in a random way.
3. it is like raindrop is falling in the bucket and you are not going to get uniform distribution both in time and space.
4. some time they may arrive faster sometime slower and due to it there is some noise is introduced in image the i.e. photon shot noise.



One can model this shot noise by Poisson distribution, here the mean value that we are trying to get. This mean value is nothing but the **brightness of the scene point**.

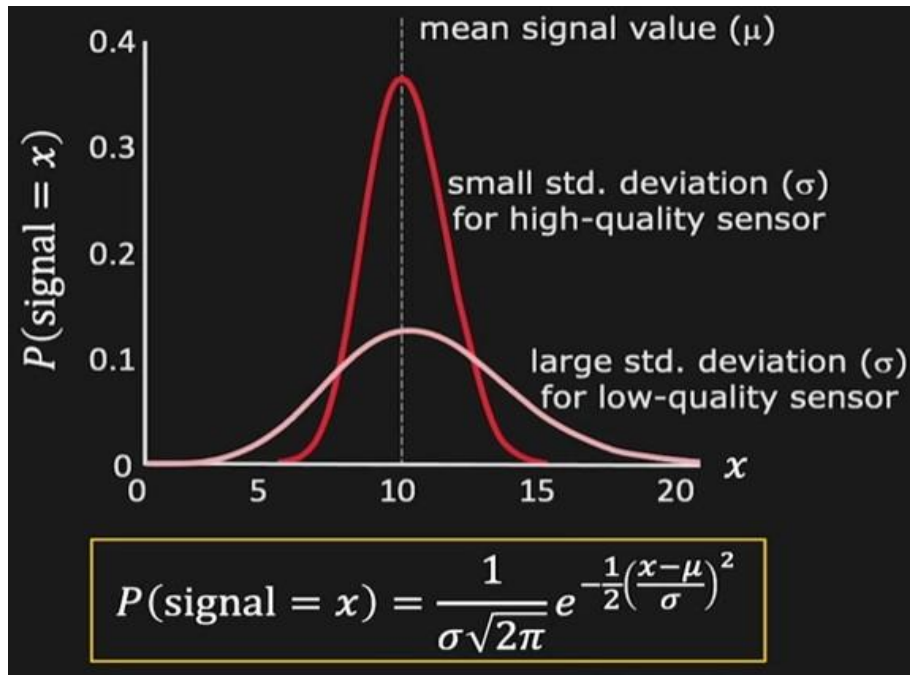
Denoising: Different Noise and their denoising approaches

Readout Noise (Scene Independent noise)

Electronic Noise

Now, photons are arrived, they need to convert into electron. Now we need to read it out using technique like CCD or CMOS.

Now, you want to convert electron into voltage. During that conversion process that circuit itself introduced noise i.e. before analog to digital conversion.



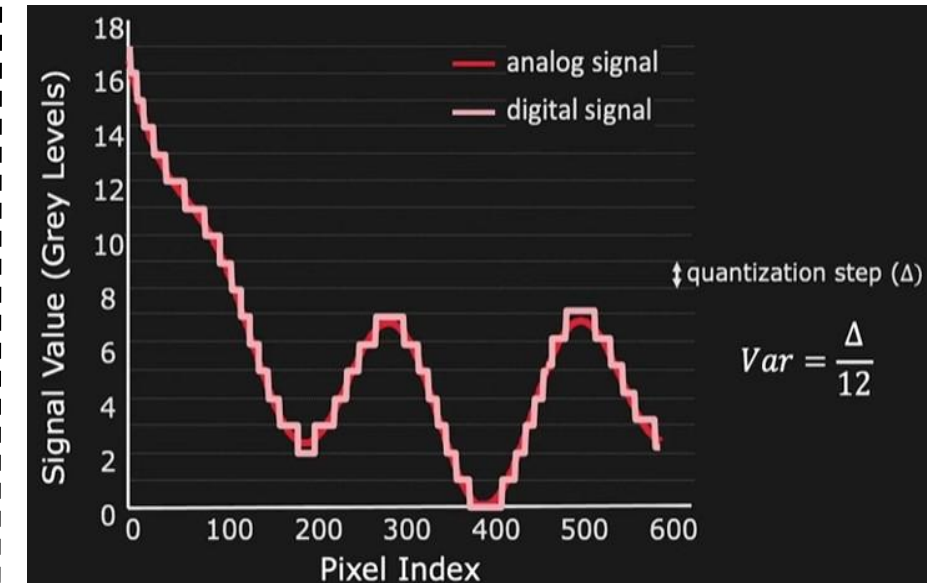
This noise is often modeled as gaussian noise. It has mean, that we are going to measure.

High quality sensor - less spread in distribution.

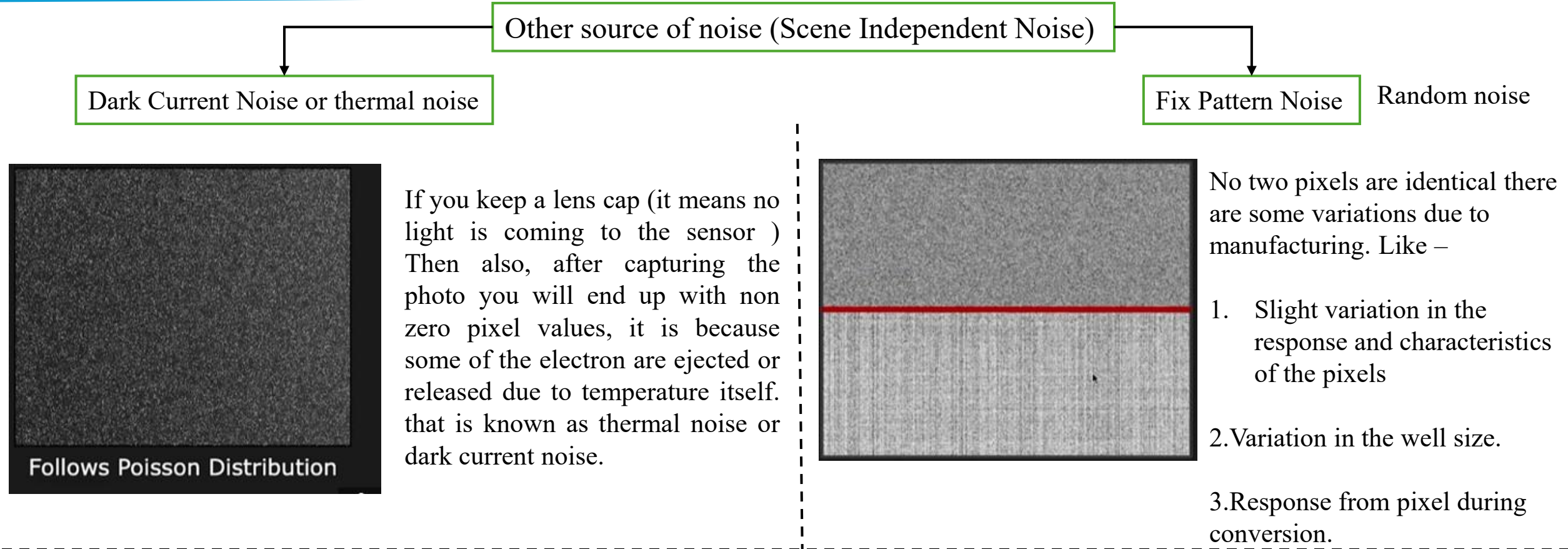
Low quality sensor – large spread in distribution.

Quantization Noise

When the camera converts the analog voltage from the sensor into a digital number (e.g., 12-bit or 14-bit), it has to "round" the value. This rounding error creates a small amount of noise.



Denoising: Different Noise and their denoising approaches



Dead Pixel Correction

The ISP identifies "outlier" pixels (Salt-and-Pepper noise) that don't match their neighbors at all. It "inpaints" these using the median value of the surrounding pixels.

Denoising: Different Noise and their denoising approaches

Different Denoising approaches -

Noise Type	Primary Source	Physical Nature	Primary Removal Strategy	Role of AI / NAFNet
1. Photon (Shot) Noise	Discrete nature of light (photons).	Random (Poisson distribution).	Temporal Averaging: Stacking 15+ frames to average out fluctuations.	Uses "Temporal Attention" to weight cleaner frames more heavily during fusion.
2.1 Electronic (Read) Noise	Sensor circuitry (Gain/Amplifiers).	Random/Gaussian but often structured.	Correlated Double Sampling (CDS): Comparing pixel voltage before and after charge transfer.	Learns to "subtract" the electronic "hiss" or floor from the signal.
2.2 Quantization Noise	Rounding errors in the ADC.	Deterministic based on Bit Depth.	Dithering or High Bit-Depth: Using 14-bit instead of 12-bit ADC.	Super-Resolution / Bit-Expansion: Reconstructing lost gradients to prevent color banding.
3. Dark Current (Thermal)	Heat-generated electrons in silicon.	Temperature-dependent and time-based.	Dark Frame Subtraction: Capturing a "black" frame with the shutter closed and subtracting it.	Estimates heat maps based on sensor metadata to clean thermal "fog."
4. Fixed Pattern Noise (FPN)	Manufacturing variations in pixels/columns.	Stationary (Static; stays in the same spot).	Flat-Field Correction: Using a factory-calibrated "gain map" to normalize pixel sensitivity.	Acts as a "Blind Inpainter" to fix "Hot Pixels" or column streaks that survive hardware filters.

White balance/ Color balance

What it is ?

In physics, different light sources have different **Spectral Power Distributions**. If you take a picture of a white piece of paper under a candle, the sensor sees orange. (**It's a Kind of environment correction.**)

What is the Goal: Adjust red, green and blue intensities such that objects that are neutral (for example white) in the real world appear neutral (white) in digital image.

HOW to achieve the goal ?

Since most Bayer sensors have twice as many Green pixels, the ISP usually treats **Green as the anchor (Gain = 1.0)** and adjusts Red and Blue to match.

1.Calculate Gains: The ISP determines multipliers (G_r and G_b).

2.Multiply: Every Red pixel is multiplied by G_r , and every Blue pixel by G_b .

3.Result: The "orange" or "blue" tint disappears, and the image looks "natural."

White balance/ Color balance

Methods to determine the Gain.

Manual / Preset White Balance-

- The Method: The user tells the camera exactly what the light is (e.g., "Tungsten" or "Daylight").
- How it works: The ISP uses a pre-calibrated **Lookup Table (LUT)** with fixed gains for that specific light source.
- Use Case: Professional studio photography where lighting never changes.

Auto White Balance (AWB) / AI-Based

- The Method: This is what modern smartphones (and likely your DRIFT pipeline) use.
- How it works: A Neural Network (often a small CNN) looks at the entire scene. It recognizes objects (like "skin," "sky," or "grass").
- The Logic: If the AI sees a "cloud," it knows a cloud *should* be white. It calculates the gains based on that specific object rather than the whole average.

Demosaicing:

Image sensors are naturally colorblind (they only measure the **intensity of light** not its wavelength).

Mosaic - Each individual pixel (photosensor) on the sensor is covered by a tiny color filter, making it sensitive to only **one** of the primary colors: Red, Green, or Blue (RGB).

Color filter array (CFA) is mosaic of tiny filters placed over the individual pixel of image sensors and filter the light into R,G,B.

The most common CFA is **Bayer Pattern** developed by (Bayer 1976).

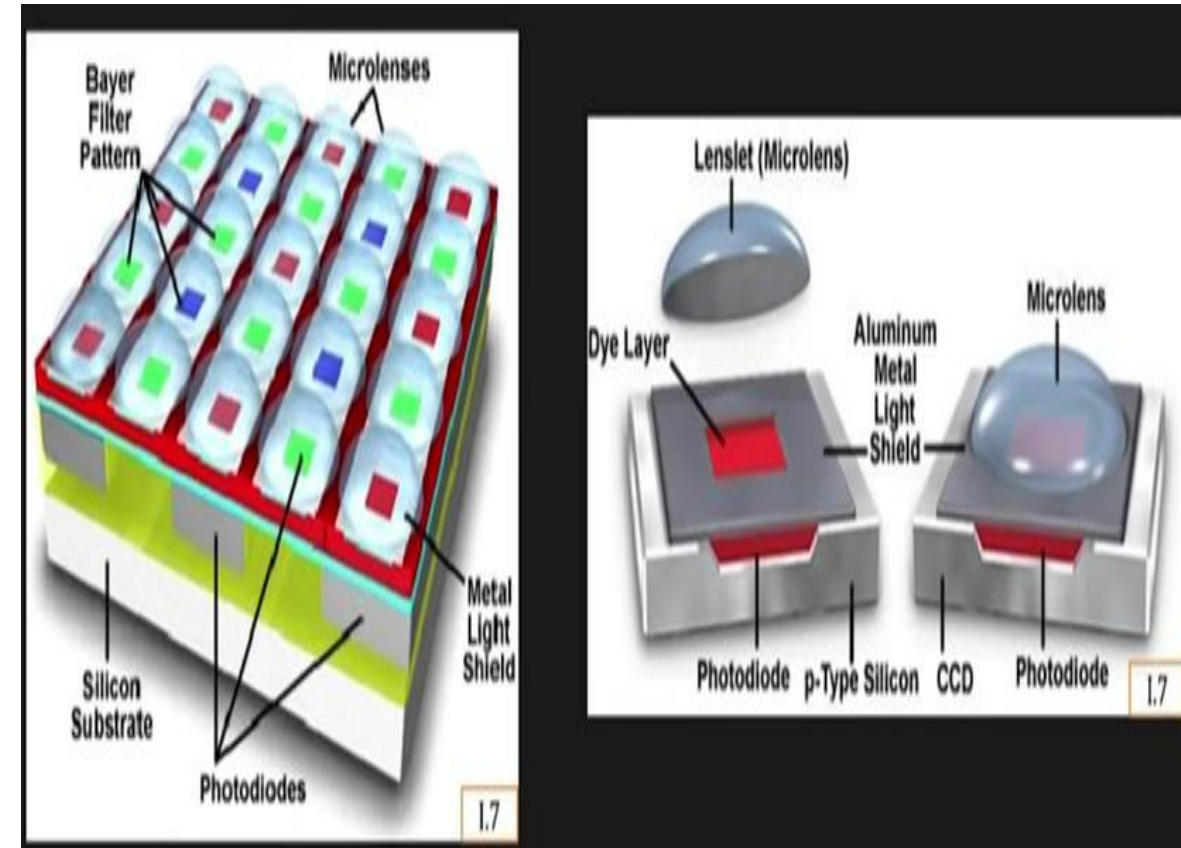
It places green filters over half of the sensors and red, blue over remaining ones.

Our human eyes are more sensitive to the green color as compared to others, this is the reason to keep twice green pixel as compared to red and blue.

or

Luminance signal (Intensity) is mostly determined by the green values, and our visual system is also more sensitive to the high frequency details in the **luminance than in chrominance**.

What is luminance and chrominance ?



Demosaicing:

Luma – Intensity or Brightness (Gamma correction always stretch luminance or intensity.) and **Chroma** - Color

Luminance is the "Brightness" or "Luma" of the image. If you remove all color from a photo, you are left with the Luminance.

Calculation of luminance: $Y = 0.299 \times R + 0.587 \times G + 0.114 \times B$

Luminance carries the **sharpness**. When you see the texture of a leaf, the fine lines of a building, or the edges of text, your brain is processing high-frequency changes in **brightness**, not color. It means luminance carries high frequency details.

Chrominance represents the "Color" or "Chroma" information.

Low Spatial Sensitivity: Human vision is "low-pass" for color. We can tell that a large area is "red," but we aren't very good at seeing sharp, tiny color details.

The Difference Signals: We calculate chrominance by subtracting the luminance from the Blue and Red channels.

Chroma blue: $C_b = B - Y$

Chroma red: $C_r = R - Y$

Why is subtraction required ?

In an RGB image, the Red, Green, and Blue channels are highly **correlated**. This means if a pixel is bright, usually all three (R, G, and B) are high. Storing them separately is a waste of space.

By subtracting Luminance (Y), we are essentially asking: *"How much does this specific color differ from the overall brightness?"*

But where is Chroma green C_g ?

We don't need C_g , because Y is already mostly made of Green (approx 59%). If we know the total brightness (Y), the Blue difference (C_b), and the Red difference (C_r), we can mathematically calculate the Green value without needing a third color channel.

Algebraic Trick: $Y - (R+B) = \text{Green}$. This saves a massive amount of data!

Demosaicing:

Each pixel on the sensor only records one color (e.g., a "red" pixel only knows how much red light it hit but has no idea about the green or blue), the raw output of the sensor **looks like a dark, speckled mosaic**.

The process of interpolating missing color values so that we have valid RGB values for all the pixels is known as **Demosaicing**.

G	R	G	R
B	G	B	G
G	R	G	R
B	G	B	G

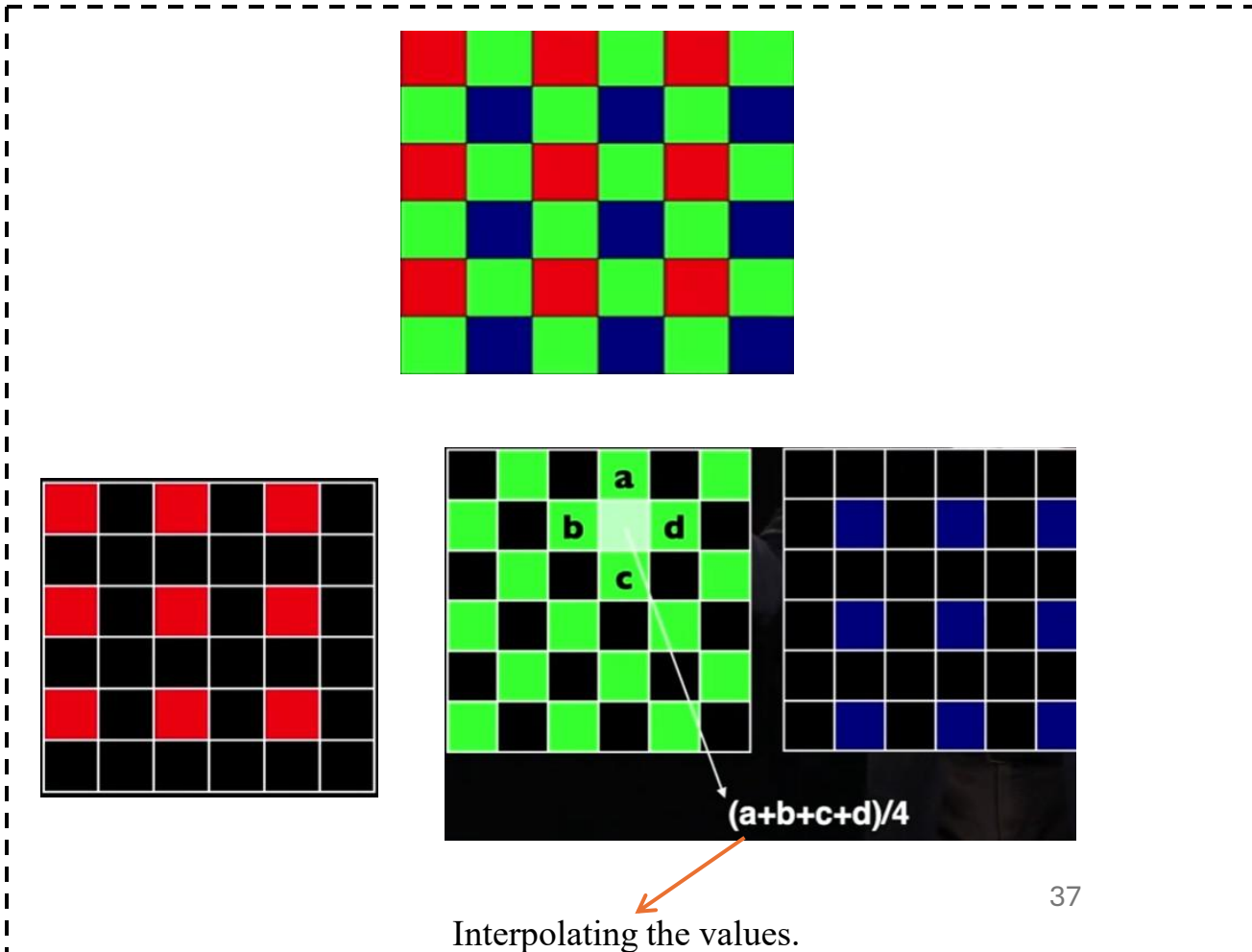
(a)

Color Filter Array (CFA) layout

rGb	Rgb	rGb	Rgb
rgB	rGb	rgB	rGb
rGb	Rgb	rGb	Rgb
rgB	rGb	rgB	rGb

(b)

Interpolated pixel values with unknown (guessed) values shown as lower case.



Interpolation:

Scale the input image (2 x 2) to (4 x 4) by using **nearest neighbor interpolation**.

Input Image

10	20
30	40



10			20
30			40

Output after coordinate transformation

Find out the intensity values for these positions



10			20
30			40

Pixel Positions for reference

1,1	1,2	1,3	1,4
2,1	2,2	2,3	2,4
3,1	3,2	3,3	3,3
4,1	4,2	4,3	4,4

Interpolation:

Output after coordinate transformation

P1 10 (1,1)	U1 (1,2)		P2 20(1,4) U3
U2			
P3 30(4,1)		U4	P4 40(4,4)

For a missing color, it simply copies the value from the closest adjacent pixel of that color.

Pixel Positions for reference

1,1	1,2	1,3	1,4
2,1	2,2	2,3	2,4
3,1	3,2	3,3	3,3
4,1	4,2	4,3	4,4

Find the Euclidean Distance for U1 from all the known points: Here D1 is distance between 10 and unkown point U1 etc.

$$P1:D1 = \text{sqrt}((1-1)^2 + (2-1)^2) = 1$$

$$P2:D2 = \text{sqrt}((1-1)^2 + (4-2)^2) = 2$$

$$P3:D3 = \text{sqrt}((4-1)^2 + (2-1)^2) = 3$$

$$P4:D4 = \text{sqrt}((4-1)^2 + (4-2)^2) = 3.6055$$

Here D1 is minimum, out of all of the distance then U1 will get intensity value = 10.

All the intensities values.

10	10	20	20
10	10	20	20
30	30	40	40
30	30	40	40

Interpolation:

Method	How it Works	Quality / Artifacts
Nearest Neighbor	For a missing color, it simply copies the value from the closest adjacent pixel of that color.	Very Poor. Causes "blockiness" and severe "zipper" artifacts along edges.
Bilinear Interpolation	Takes the average of the 2 or 4 surrounding pixels of the same color.	Average. Much smoother than Nearest Neighbor, but it "blurs" sharp edges and fine details.
Bicubic Interpolation	Uses a weighted average of a 4 x 4 neighborhood (16 pixels).	Good. Sharper and smoother than bilinear, but computationally more expensive.

Color correction matrix (CCM) - Hardware Correction

- The "Why":**

No two sensors see "Red" the same way.

The chemical makeup of the micro-lenses and the color filters on the CMOS chip determines how the sensor interprets the spectrum.

Purpose: Sensors don't see "Red" exactly like a standard monitor (sRGB). The CCM maps the sensor's specific color space to a standard one.

- The "How":**

It uses that 3 x 3 matrix to "cross-talk" the colors. (Basically, it perform the matrix multiplication which is a **linear operation**.)

It says: *"To get a true sRGB Red, I need 110% of the sensor's Red minus 10% of its Green."*

•**The Result:** It ensures that "Apple Red" looks like "Apple Red" regardless of which company manufactured the sensor chip.

$$\begin{bmatrix} R_{out} \\ G_{out} \\ B_{out} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} R_{in} \\ G_{in} \\ B_{in} \end{bmatrix}$$

Color correction matrix (CCM) - Hardware Correction



Macbeth Color Checker Chart

They use a **Macbeth ColorChecker** (a chart with 24 specific, scientifically calibrated color squares).

Steps followed by algorithm -

- 1.Capture:** Take a RAW photo of the chart under a specific light (e.g., D65/Daylight).
- 2.Measure:** Compare the sensor's raw output for each square to the "Ground Truth" values of that chart.
- 3.Optimization:** Use a least-squares algorithm to find the 9 numbers (a_{11} to a_{33}) that minimize the error across all 24 colors.

Note: White balance : Environment Correction
CCM : Hardware Correction

Gamma Correction: Global Correction

Gamma Correction (The "Human Eye" Block)

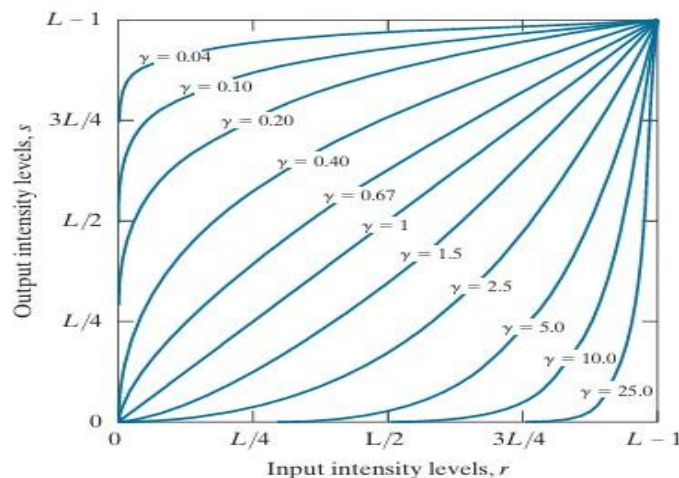
The Responsibility: Digital sensors are **Linear** (if you double the light, the pixel value doubles). However, the human eye is **Non-Linear**—we are much more sensitive to changes in dark shadows than in bright highlights.

•**The Action:** This block applies a mathematical curve to "squash" the highlights and "stretch" the shadows so the image looks natural to a human.

To rectify the same, one can utilize power law transformation or Gamma Corrections. Basic idea is to raise the pixel value with certain power to improve the overall brightness and contrast of the image.

$$S = c r^{\gamma},$$

Where - r = Input Intensity Level, S = Output Intensity Level, C = constant., γ = gamma value



$\gamma < 1$ = Overall image gets brighter.

$\gamma > 1$ = Overall image gets darker.

$\gamma = 1$ = Identity or no change in the image.

Gamma Correction:

Case 1: Dark Pixel ($r = 7$) with $\gamma = 0.5$

$\gamma < 1$, we expect this pixel to become significantly **brighter**.

1.Normalize: $7 / 255 = 0.02745$

2.Apply Gamma: $(0.02745)^{0.5} = 0.16568$

3.Scale back: $0.16568 \times 255 = 42.25$

Result: The pixel value jumps from **7 to 42**. This "stretches" the dark detail, making it much more visible.

Case 2: Bright Pixel ($r = 240$) with $\gamma = 1.5$

$\gamma > 1$, we expect this pixel to become **darker**.

1.Normalize: $(240 / 255) = 0.94118$

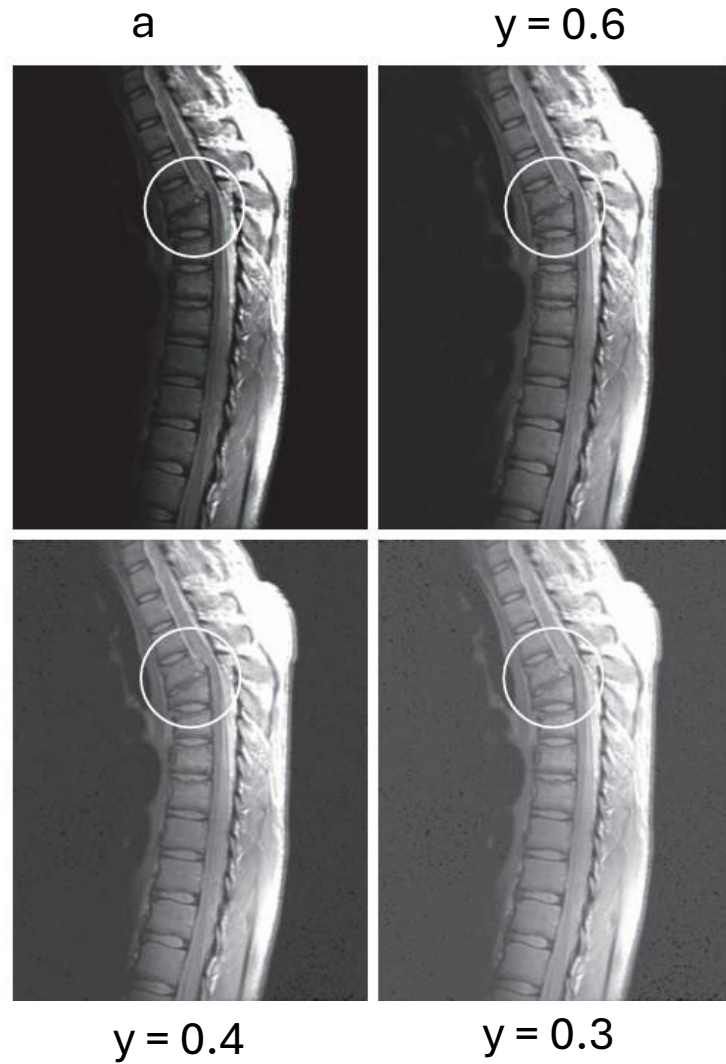
2.Apply Gamma: $(0.94118)^{1.5} = 0.91354$

3.Scale back: $0.91354 \times 255 = 232.95$

Result: The pixel value drops from **240 to 233**. Even though it is a small numerical drop, in an image, this helps reduce "blowout" and adds depth to highlight areas.

In image processing, we almost always prefer **Gamma Corrections** over Log/Exponential because of its **tunability by varying the gamma value**.

Gamma Correction:



(a) MRI of fractured human spine

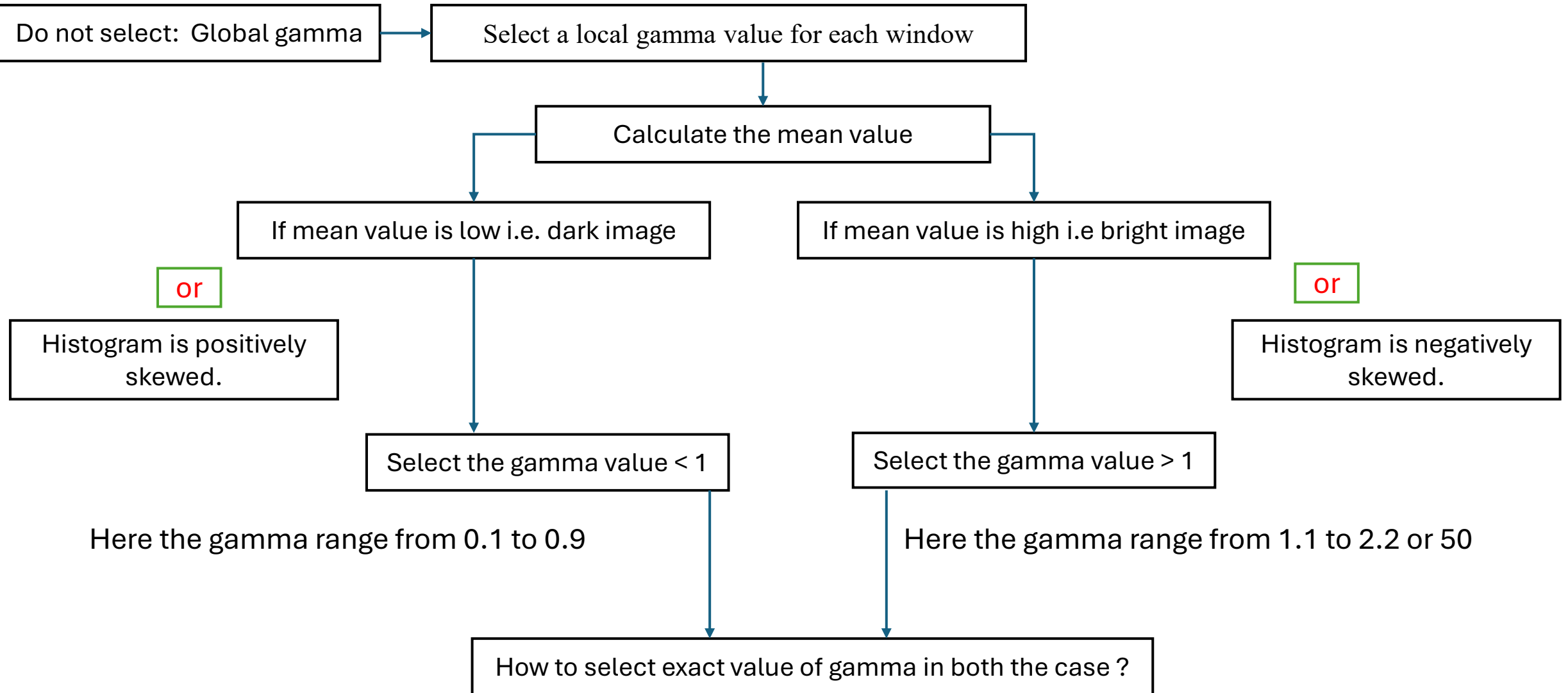
As gamma decreases then more details are visible in the dark area.



(a1) Aerial image of airport runway

As gamma increasing's then more details are visible in the bright area.

Local Gamma Correction or Adaptive Gamma correction.



Note: In most of the TV monitors standard gamma value is 2.2, it helps to see a lot of details.

Local Gamma Correction or Adaptive Gamma correction.

How to select exact value of gamma in both the case ?

$$C = r^y$$

Gamma correction formula

$$\ln(c) = y \ln(r)$$

Take log on both side.

$$Y = \ln(c) / \ln(r)$$

Here c is mid grey level range and m is mean value
Calculated in each local window.

$$Y = \ln(0.5) / \ln(m/255)$$



At the end, Apply the gaussian blur on the gamma map
to avoid the blocky or patchy appearance.

Tone Mapping (Local)

What it is ?

Tone Mapping is the process of mapping a **High Dynamic Range (HDR)** image—where the difference between the darkest and brightest parts is massive—into a **Standard Dynamic Range (SDR)** that can be displayed on a screen (like your phone or monitor).

- **The Input:** A 16-bit or 32-bit "Linear RAW" image.
- **The Output:** An 8-bit or 10-bit image that fits the "bottleneck" of current display technology.

Why is it required ?

The real world has an incredible range of brightness. A bright sun is millions of times brighter than a dark shadow.

- **Human Eye:** Can perceive about **20 to 24 stops** of dynamic range by adjusting the pupil and processing in the brain.
- **Modern Sensors:** Can capture about **12 to 14 stops** (more with your 15-frame burst).
- **Standard Displays (SDR):** Can only show about **6 to 8 stops**.

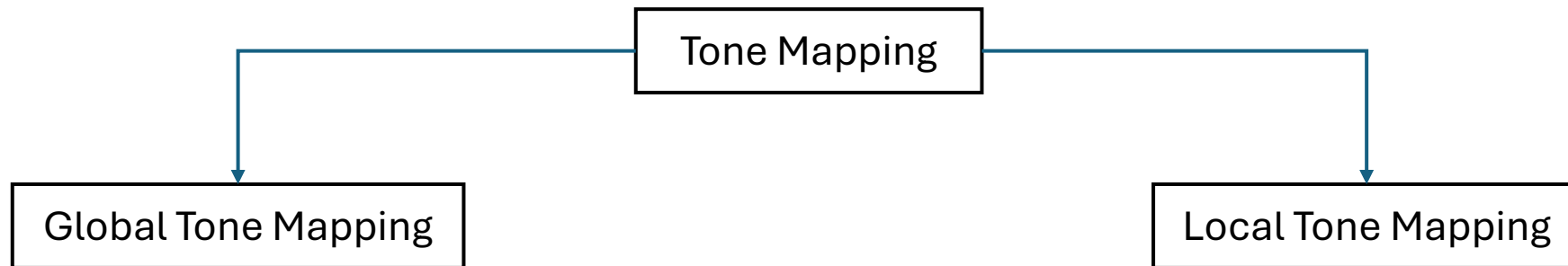
Tone Mapping (Local)

Purpose of Tone Mapping

The primary goal is to **compress the range while maintaining local contrast**. If you just "shrink" everything (Global Tone Mapping), the image looks "flat" and gray.

Good Tone Mapping (Local/Adaptive) ensures that:

- The sky stays blue and detailed.
- The person in the shadow is visible and bright.
- The edges of objects still look sharp.



It applies the **same math curve** to every pixel in the image.

It analyzes the image in **blocks** or **regions**.
It calculates a different brightness adjustment for every part of the photo.

Edge Enhancement (Unsharp Masking) and High Boost Filtering

What it is: Applying filters (like Unsharp Masking).

Purpose: Since Denoising and Demosaicing tend to "soften" the image, this step adds high-frequency detail back to edges to make the image look crisp.

A technique which helps to sharp the image without using image sharpening filter.

Steps

1. Blur the original image.
2. Subtract the blurred image from the original image.
3. Add the mask to the original image.

$$g_{\text{mask}}(x, y) = f(x, y) - \bar{f}(x, y)$$

$$g(x, y) = f(x, y) + k g_{\text{mask}}(x, y)$$

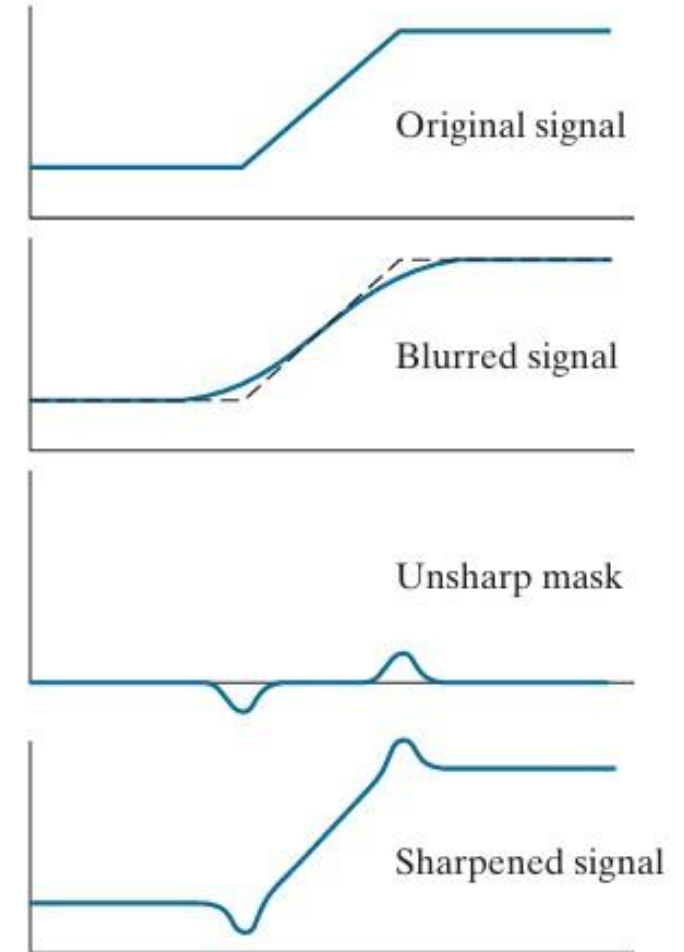
Where –

$f(x, y)$ = original image

$\bar{f}(x, y)$ = blurred image

g_{mask} = unsharp mask

Note: Consider a situation where sharpening filter is not available



Unsharp masking and highboost filtering

Edge Enhancement (Unsharp Masking) and High Boost Filtering

Apply unsharp masking $k = 1$ and high boost filtering with $k = 4$.

Original Image

6	7	8
3	4	5
6	8	4

Padded Image

6	6	7	8	8
6	6	7	8	8
3	3	4	5	5
6	6	8	4	4
6	6	8	4	4

Note: Padding is done by duplicating row and columns

Gaussian Kernel with $\sigma = 3$

0.1069	0.1131	0.1069
0.1131	0.1195	0.1131
0.1069	0.1131	0.1069

Steps for reference

Steps

1. Blur the original image.
2. Subtract the blurred image from the original image.
3. Add the mask to the original image.

$$g_{\text{mask}}(x, y) = f(x, y) - \bar{f}(x, y)$$

$$g(x, y) = f(x, y) + k g_{\text{mask}}(x, y)$$

Edge Enhancement (Unsharp Masking) and High Boost Filtering

Blurred Image

5.3432	6.0158	6.8884
5.3937	5.6449	5.8589
5.5632	5.3684	5.098

Approximate value of Blurred Image

5	6	7
5	6	6
6	5	5

Mask

1	1	1
2	2	1
0	3	1

=

Original Image

6	7	8
3	4	5
6	8	4

Subtract

Approximate value of Blurred Image

5	6	7
5	6	6
6	5	5

Output: Unsharp Masking (K=1)

7	8	9
5	6	6
6	11	5

=

Original Image

6	7	8
3	4	5
6	8	4

+

K x

Mask

1	1	1
2	2	1
0	3	1

Output: High boost Filtering (K=4)

10	11	12
11	12	9
6	20	8

=

6	7	8
3	4	5
6	8	4

+

K x

Mask

1	1	1
2	2	1
0	3	1

Edge Enhancement (Unsharp Masking) and High Boost Filtering

Original image



Image blurred using 31 x 31 LPF



Mask



Unsharp masking with $k = 1$



High boost filtering with $k = 4.5$

Color Space Conversion (RGB to YCbCr)

Purpose: Preparing the image for the human eye's sensitivity.

What it is: Splitting the image into Y (Luminance) and Cb, Cr (Chrominance).

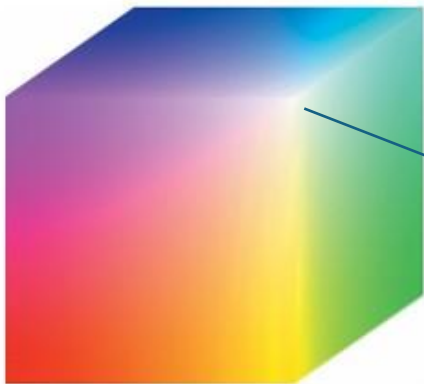
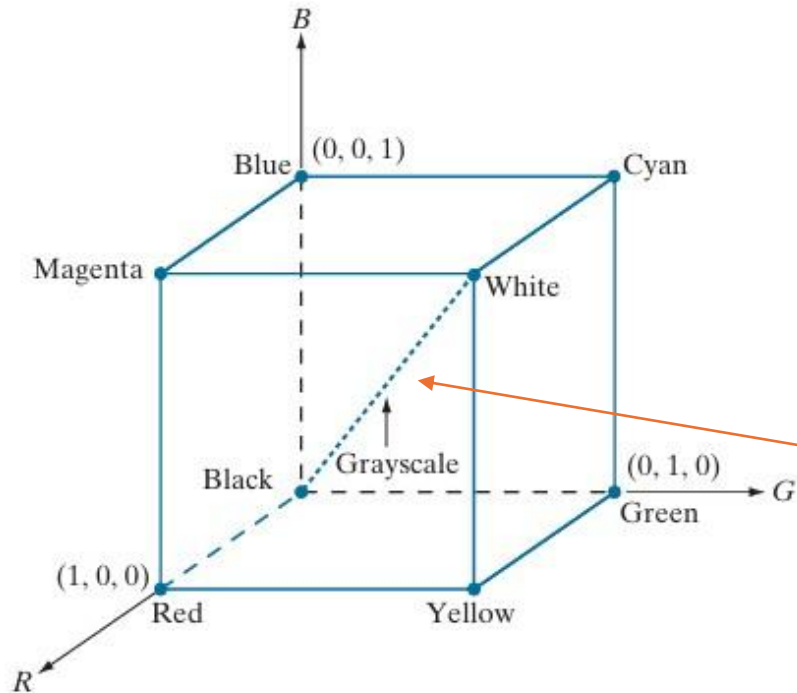
$$\begin{bmatrix} Y \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.1687 & -0.3313 & 0.5 \\ 0.5 & -0.4187 & -0.0813 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 0.5 \\ 0.5 \end{bmatrix}$$

Various steps for RGB to YCbCr Conversion -

- 1.Normalize:** Convert your RGB pixels from integers (0–255) to floats (0.0–1.0).
- 2.Apply Matrix:** Multiply each pixel's RGB vector by the transformation matrix.
- 3.Offset:** Add the 128 (for 8-bit) or 0.5 (for float) bias to the Cb and Cr channels so that "neutral gray" is in the middle.
- 4.Subsample (Optional):** If you are mimicking JPEG compression, you would now "shrink" the Cb and Cr layers while keeping Y at full size.

Color Model: RGB (Red-Green-Blue)

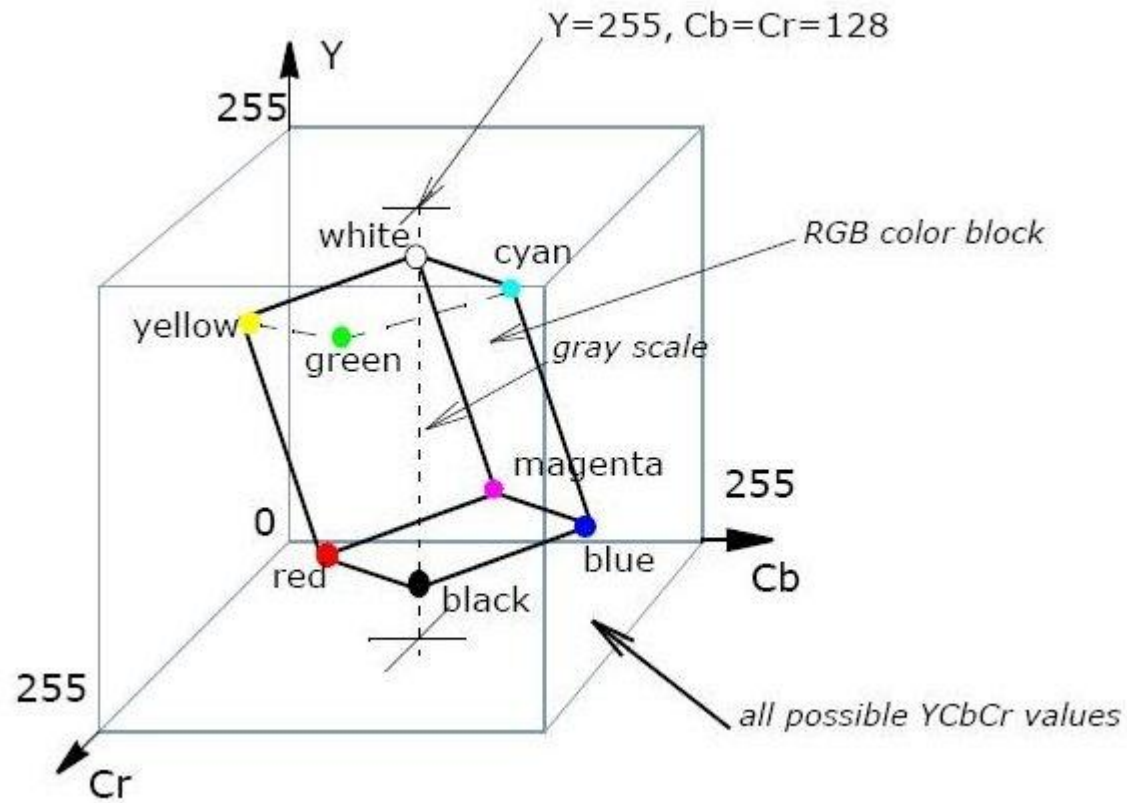
Color Model RGB



RGB Color
Cube

1. RGB model based on cartesian coordinate system.
2. Primary colors R,G, and B are at the three corners.
3. Secondary colors like Cyan, Magenta and yellow at the other corners.
4. Black at the origin and white at the corner which is farthest away from the origin.
5. Grey scale extend from black to white along the line joining these two points.
6. Assumption is that all R,G and B values are normalized between 0 to 1.
7. Pixel Depth – No. of bits used to represent each pixel in RGB space. Each RGB color pixel has a depth of 24bits.
8. let's say it's an 8-bit image, then the limits of cube along each axis becomes [0,255].

YcbCr : Luma- Blue difference Chroma- Red difference chroma



It follows a Cartesian coordinate system. It's a three-dimensional system where three component Y, Cb, Cr are treated as three orthogonal components.

It is most widely used in digital video and photography systems, including DVD, Blue-ray disc, digital TV and JPEG images.

Y (Luma) - It represent the brightness or luminance of the image.

Cb (Blue difference chroma) – It represent the difference between blue color and the luma.

Cr (Red difference chroma) – it is difference between red color and chroma.

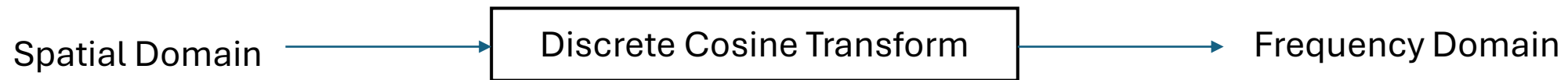
YCbCr – it allows better compression of image and video data.

Image Compression: Discrete Cosine Transform: DCT

What it is: Chroma Subsampling and DCT (Discrete Cosine Transform).

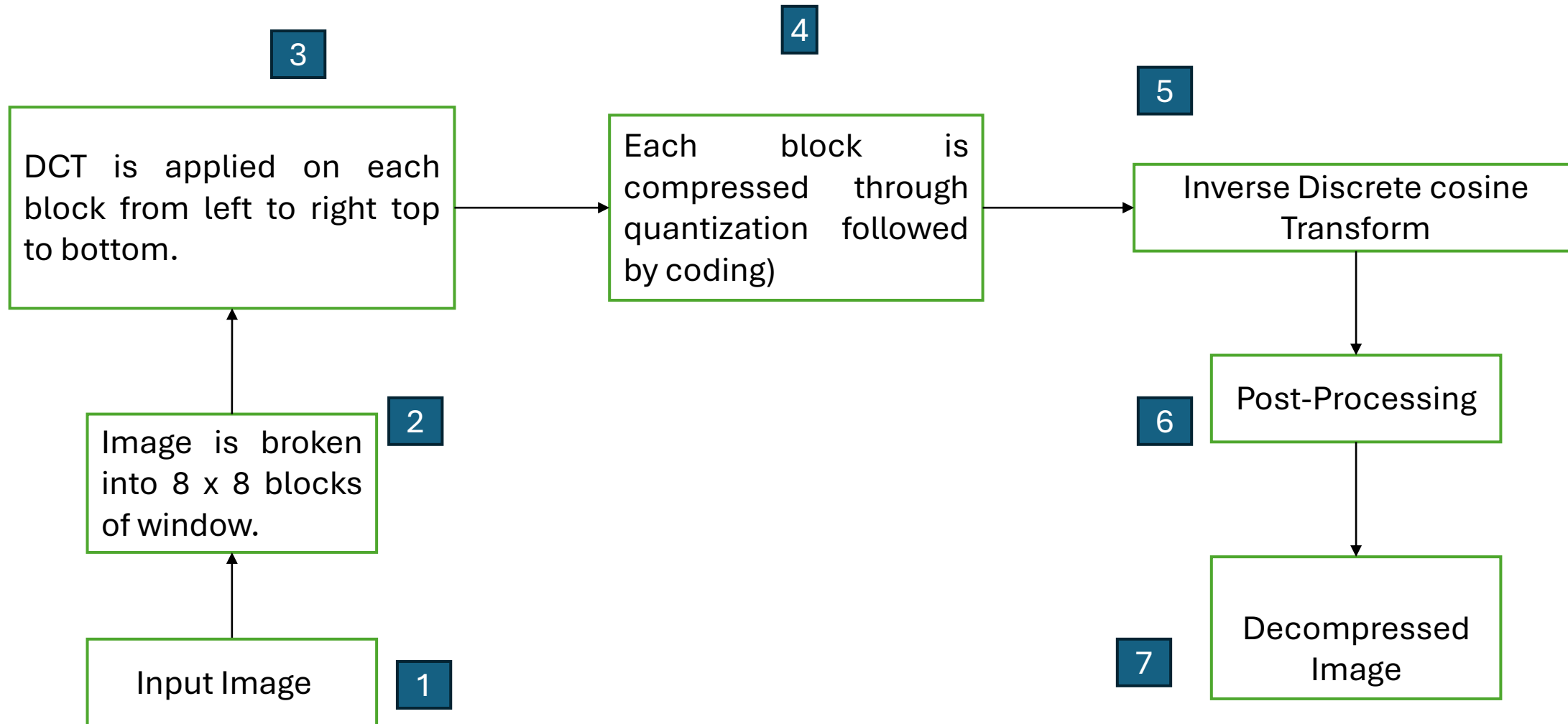
Purpose: Reducing the file size for storage.

Lossy Compression – In this type of compression, after performing the inverse transformation we can't get exactly the same image as the older one (i.e. Loss of information). Overall the quality of the image gets significantly reduced.

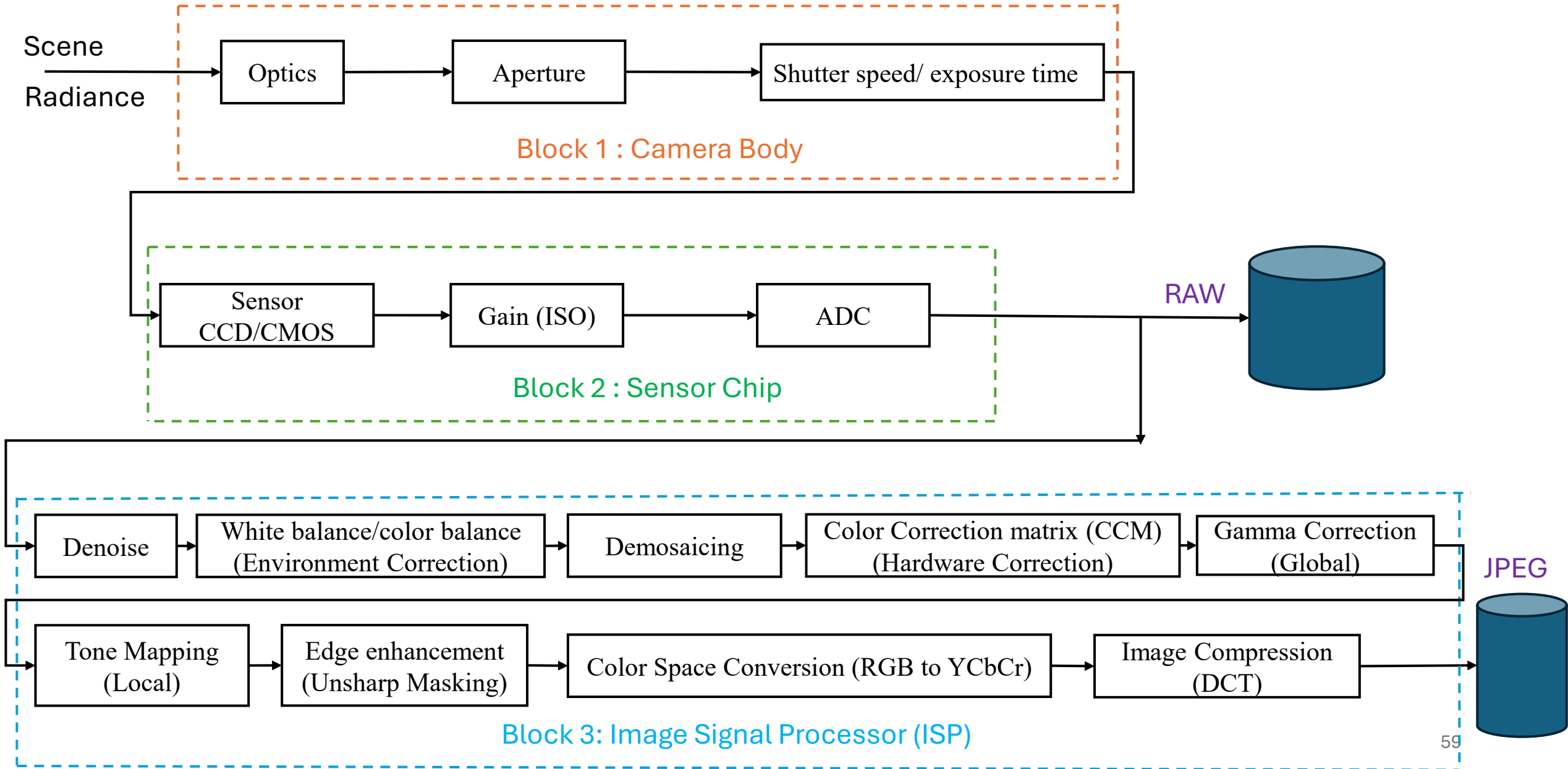


1. Discrete Cosine Transform (DCT) is a lossy image compression technique.
2. DCT represents an image as a sum of sinusoids of varying magnitudes and frequencies.
3. In DCT, most of the significant information or signal (image) energy, is concentrated in the **few number of coefficients (near to the origin) and rest other frequency having very small information** which can be stored by using less number of bits.
4. DCT coefficients are **real valued** while DFT coefficients are complex, therefore, hardware implementation of DCT is easier than DFT.

Lossy Compression: DCT : Flow Chart



Camera Image Sensing Pipeline: (Camera ISP) Overall flow diagram



End Term Question Patterns

Total 3 sections –

Section A – 6Qx 2M = 12 Marks. (Theory)

Total 8 questions.

Attempt any 6 questions.

Each question carries 2 marks.

Section B – 2Q x 6M = 12 Marks. (Theory + justification of concepts)

Total 3 questions.

Attempt any 2 questions.

Each question carries 6 marks.

Section C – 2Q x 8M = 16 Marks. (Numerical)

Total 2 questions.

Each question carries 8 marks.

There will be internal choice in each question.